

# ■ CLASS IV ENGLISH QUESTION BANK

## CHAPTER 11 QUESTION BANK (250 QUESTIONS)

|                                                     |                                                               |
|-----------------------------------------------------|---------------------------------------------------------------|
| <b>Subject:</b> English                             | <b>Class:</b> IV                                              |
| <b>Chapter:</b> 11                                  | <b>Total Questions:</b> 250 (25 per category across 10 files) |
| <b>Source:</b> Praxis Comprehension & Composition 4 | <b>Board / Blueprint:</b> ICSE Class IV English Pattern       |

### 1. Multiple Choice Questions (MCQs)

#### Question 1

Question ID: CH11-MCQ-001

Type: MCQ

Difficulty: Easy

Bloom Level: Remember

Topic: Number of Life Stages

Marks: 1

**Question:** How many stages of life does a frog go through during its lifecycle?

- (A) Two stages
- (B) Three stages
- (C) Four stages
- (D) Five stages

#### Question 2

Question ID: CH11-MCQ-002

Type: MCQ

Difficulty: Easy

Bloom Level: Remember

Topic: Definition of Metamorphosis

Marks: 1

**Question:** What is the process of changing from one form to another as a frog grows called?

- (A) Hibernation
- (B) Metamorphosis
- (C) Migration
- (D) Photosynthesis

#### Question 3

Question ID: CH11-MCQ-003

Type: MCQ

Difficulty: Easy

Bloom Level: Remember

Topic: First Life Stage

Marks: 1

**Question:** What is the first stage in a frog's lifecycle?

- (A) Tadpole
- (B) Froglet
- (C) Egg
- (D) Adult Frog

#### Question 4

Question ID: CH11-MCQ-004

Type: MCQ

Difficulty: Easy

Bloom Level: Remember

Topic: Egg Laying (Spawning)

Marks: 1

**Question:** Who lays the eggs in water to start a new lifecycle?

- (A) Male frogs
- (B) Female frogs
- (C) Tadpoles
- (D) Froglets

#### Question 5

Question ID: CH11-MCQ-005

Type: MCQ

Difficulty: Easy

Bloom Level: Remember

Topic: Egg Hatching Duration

Marks: 1

**Question:** How long does it take for frog eggs to hatch?

- (A) 1 to 3 weeks
- (B) 4 to 6 weeks
- (C) 2 to 5 months
- (D) 1 to 3 days

#### Question 6

Question ID: CH11-MCQ-006

Type: MCQ

Difficulty: Easy

Bloom Level: Remember

Topic: Creature Hatching from Egg

Marks: 1

**Question:** What hatches out of a frog egg?

- (A) Adult Frog
- (B) Tadpole
- (C) Fish
- (D) Caterpillar

### Question 7

Question ID: CH11-MCQ-007

Type: MCQ

Difficulty: Easy

Bloom Level: Remember

Topic: Tadpole Body Features

Marks: 1

**Question:** A tadpole is born with a \_\_\_\_\_ but no \_\_\_\_\_.

- (A) legs; tail
- (B) tail; legs
- (C) lungs; gills
- (D) wings; tail

### Question 8

Question ID: CH11-MCQ-008

Type: MCQ

Difficulty: Easy

Bloom Level: Remember

Topic: Tadpole Habitat

Marks: 1

**Question:** Where does a tadpole live?

- (A) On land
- (B) In water
- (C) In trees
- (D) Under rocks in dry desert

### Question 9

Question ID: CH11-MCQ-009

Type: MCQ

Difficulty: Easy

Bloom Level: Remember

Topic: Tadpole Breathing Organ

Marks: 1

**Question:** How does a tadpole breathe while living in water?

- (A) With lungs

- (B) With gills
- (C) With nose
- (D) With tail

### Question 10

Question ID: CH11-MCQ-010

Type: MCQ

Difficulty: Easy

Bloom Level: Remember

Topic: Third Life Stage (Froglet)

Marks: 1

**Question:** A tadpole grows into which intermediate stage?

- (A) Egg
- (B) Froglet
- (C) Caterpillar
- (D) Fish

### Question 11

Question ID: CH11-MCQ-011

Type: MCQ

Difficulty: Easy

Bloom Level: Remember

Topic: Froglet Breathing Organ

Marks: 1

**Question:** What organ develops in a froglet that helps it breathe air?

- (A) Gills
- (B) Lungs
- (C) Fins
- (D) Tail

### Question 12

Question ID: CH11-MCQ-012

Type: MCQ

Difficulty: Easy

Bloom Level: Remember

Topic: Froglet Movement Adaptation

Marks: 1

**Question:** What physical feature develops in a froglet for hopping on land?

- (A) Wings
- (B) Legs
- (C) Fins
- (D) Shell

### Question 13

Question ID: CH11-MCQ-013

Type: MCQ

Difficulty: Easy

Bloom Level: Remember

Topic: Last Life Stage

Marks: 1

**Question:** What is the last stage of a frog's lifecycle?

- (A) Tadpole
- (B) Adult Frog
- (C) Egg
- (D) Froglet

### Question 14

Question ID: CH11-MCQ-014

Type: MCQ

Difficulty: Easy

Bloom Level: Remember

Topic: Adult Frog Feature

Marks: 1

**Question:** Unlike a tadpole, an adult frog has:

- (A) A long tail
- (B) No tail
- (C) Gills only
- (D) No legs

### Question 15

Question ID: CH11-MCQ-015

Type: MCQ

Difficulty: Medium

Bloom Level: Understand

Topic: Lifecycle Restart Trigger

Marks: 1

**Question:** When does a new lifecycle of a frog begin?

- (A) When an adult frog loses its legs
- (B) When a female frog lays eggs
- (C) When a tadpole learns to fly
- (D) When winter arrives

### Question 16

Question ID: CH11-MCQ-016

Type: MCQ

Difficulty: Medium

*Bloom Level: Understand*

*Topic: Vocabulary Meaning - Metamorphosis*

*Marks: 1*

**Question:** What is the textbook meaning of **'metamorphosis'**?

- (A) The process of changing from one form to another
- (B) Coming out of an egg
- (C) Sleeping through winter
- (D) Flying in the sky

### Question 17

*Question ID: CH11-MCQ-017*

*Type: MCQ*

*Difficulty: Medium*

*Bloom Level: Understand*

*Topic: Vocabulary Meaning - Hatch*

*Marks: 1*

**Question:** What is the textbook meaning of **'hatch'**?

- (A) Coming out of egg
- (B) Laying eggs in water
- (C) Hopping on land
- (D) Losing legs

### Question 18

*Question ID: CH11-MCQ-018*

*Type: MCQ*

*Difficulty: Medium*

*Bloom Level: Understand*

*Topic: Difference - Tadpole vs Adult Frog*

*Marks: 1*

**Question:** A tadpole lives in \_\_\_\_\_ and has a \_\_\_\_\_, while an adult frog can live on land and has \_\_\_\_\_ tail.

- (A) air; legs; long
- (B) water; tail; no
- (C) trees; shell; short
- (D) soil; wing; red

### Question 19

*Question ID: CH11-MCQ-019*

*Type: MCQ*

*Difficulty: Hard*

*Bloom Level: Evaluate*

*Topic: Biological Classification*

*Marks: 1*

**Question:** Animals like frogs that can live both in water (as tadpoles) and on land (as adults) are called what?

- (A) Reptiles
- (B) Amphibians
- (C) Mammals
- (D) Birds

### Question 20

Question ID: CH11-MCQ-020

Type: MCQ

Difficulty: Medium

Bloom Level: Understand

Topic: Sequential Stages Order

Marks: 1

**Question:** Which sequence correctly orders the four stages of a frog's lifecycle?

- (A) Egg → Tadpole → Froglet → Adult Frog
- (B) Tadpole → Egg → Adult Frog → Froglet
- (C) Adult Frog → Froglet → Tadpole → Egg
- (D) Egg → Froglet → Tadpole → Adult Frog

### Question 21

Question ID: CH11-MCQ-021

Type: MCQ

Difficulty: Medium

Bloom Level: Apply

Topic: Grammar - Plural Noun

Marks: 1

**Question:** What is the plural form of the noun 'tadpole'?

- (A) Tadpoles
- (B) Tadpoless
- (C) Tadpolies
- (D) Tadpolen

### Question 22

Question ID: CH11-MCQ-022

Type: MCQ

Difficulty: Medium

Bloom Level: Apply

Topic: Grammar - Action Verb

Marks: 1

**Question:** In the sentence 'A froglet has legs for hopping', what part of speech is 'hopping'?

- (A) Noun
- (B) Verb / Participle action

- (C) Conjunction
- (D) Preposition

### Question 23

Question ID: CH11-MCQ-023

Type: MCQ

Difficulty: Medium

Bloom Level: Apply

Topic: Preposition Choice

Marks: 1

**Question:** Fill in the blank: \*"A tadpole grows \_\_\_\_\_ a froglet."\*

- (A) into
- (B) under
- (C) between
- (D) off

### Question 24

Question ID: CH11-MCQ-024

Type: MCQ

Difficulty: Hard

Bloom Level: Analyze

Topic: Biological Adaptation Purpose

Marks: 1

**Question:** Why is developing lungs essential for the froglet stage?

- (A) To swim faster in deep ocean water.
- (B) To allow the frog to breathe atmospheric oxygen when moving from water onto land.
- (C) To lay eggs.
- (D) To change skin color.

### Question 25

Question ID: CH11-MCQ-025

Type: MCQ

Difficulty: Easy

Bloom Level: Remember

Topic: Term for Egg Laying

Marks: 1

**Question:** What is the process of female frogs laying egg masses in water called?

- (A) Spawning
- (B) Roosting
- (C) Nesting
- (D) Molting

## Answer Key



- | \*\*CH11-MCQ-001\*\* | \*\*(C)\*\* | Page 48 states: Four stages of life. |
- | \*\*CH11-MCQ-002\*\* | \*\*(B)\*\* | Metamorphosis: process of changing form. |
- | \*\*CH11-MCQ-003\*\* | \*\*(C)\*\* | Egg is the first stage. |
- | \*\*CH11-MCQ-004\*\* | \*\*(B)\*\* | Female frogs lay eggs. |
- | \*\*CH11-MCQ-005\*\* | \*\*(A)\*\* | Takes 1 to 3 weeks to hatch. |
- | \*\*CH11-MCQ-006\*\* | \*\*(B)\*\* | Tadpoles hatch from eggs. |
- | \*\*CH11-MCQ-007\*\* | \*\*(B)\*\* | Born with a tail but no legs. |
- | \*\*CH11-MCQ-008\*\* | \*\*(B)\*\* | Lives in water. |
- | \*\*CH11-MCQ-009\*\* | \*\*(B)\*\* | Breathes with gills in water. |
- | \*\*CH11-MCQ-010\*\* | \*\*(B)\*\* | Tadpole grows into a froglet. |
- | \*\*CH11-MCQ-011\*\* | \*\*(B)\*\* | Froglet develops lungs to breathe air. |
- | \*\*CH11-MCQ-012\*\* | \*\*(B)\*\* | Develops legs for hopping. |
- | \*\*CH11-MCQ-013\*\* | \*\*(B)\*\* | Last stage is Adult Frog. |
- | \*\*CH11-MCQ-014\*\* | \*\*(B)\*\* | Adult frog has no tail. |
- | \*\*CH11-MCQ-015\*\* | \*\*(B)\*\* | New cycle begins when female lays eggs. |
- | \*\*CH11-MCQ-016\*\* | \*\*(A)\*\* | Metamorphosis = Process of changing from one form to another. |
- | \*\*CH11-MCQ-017\*\* | \*\*(A)\*\* | Hatch = Coming out of egg. |
- | \*\*CH11-MCQ-018\*\* | \*\*(B)\*\* | Tadpole: water, tail; Adult: land, no tail. |
- | \*\*CH11-MCQ-019\*\* | \*\*(B)\*\* | Amphibians live in water and on land. |
- | \*\*CH11-MCQ-020\*\* | \*\*(A)\*\* | Egg -> Tadpole -> Froglet -> Adult Frog. |
- | \*\*CH11-MCQ-021\*\* | \*\*(A)\*\* | Tadpole + s = Tadpoles. |
- | \*\*CH11-MCQ-022\*\* | \*\*(B)\*\* | Hopping indicates action verb/gerund. |
- | \*\*CH11-MCQ-023\*\* | \*\*(A)\*\* | Preposition 'into' shows transformation. |
- | \*\*CH11-MCQ-024\*\* | \*\*(B)\*\* | Lungs enable breathing air on land. |
- | \*\*CH11-MCQ-025\*\* | \*\*(A)\*\* | Laying egg mass in water is called spawning. |

## 2. Fill in the Blanks (Fillups)

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### Question 1

Question ID: CH11-FIL-001

Type: Fill in the Blanks

Difficulty: Easy

Bloom Level: Remember

Topic: Number of Life Stages

Marks: 1

**Question:** A frog goes through \_\_\_\_\_ stages of life.

### Question 2

Question ID: CH11-FIL-002

Type: Fill in the Blanks

Difficulty: Easy

Bloom Level: Remember

Topic: Process Name

Marks: 1

**Question:** As the frog grows, it changes form; this process is called \_\_\_\_\_.

### Question 3

Question ID: CH11-FIL-003

Type: Fill in the Blanks

Difficulty: Easy

Bloom Level: Remember

Topic: Egg Laying

Marks: 1

**Question:** \_\_\_\_\_ frogs lay eggs in water.

### Question 4

Question ID: CH11-FIL-004

Type: Fill in the Blanks

Difficulty: Easy

Bloom Level: Remember

Topic: Hatching Duration

Marks: 1

**Question:** It takes from one to \_\_\_\_\_ weeks for frog eggs to hatch.

### Question 5

Question ID: CH11-FIL-005

Type: Fill in the Blanks

Difficulty: Easy

Bloom Level: Remember

Topic: Hatching Outcome

Marks: 1

**Question:** \_\_\_\_\_ hatch from the frog eggs.

### Question 6

Question ID: CH11-FIL-006

Type: Fill in the Blanks

Difficulty: Easy

Bloom Level: Remember

Topic: Tadpole Body Parts

Marks: 1

**Question:** A tadpole is born with a tail but no \_\_\_\_\_.

### Question 7

Question ID: CH11-FIL-007

Type: Fill in the Blanks

Difficulty: Easy

Bloom Level: Remember

Topic: Tadpole Habitat

Marks: 1

**Question:** A tadpole lives in \_\_\_\_\_.

### Question 8

Question ID: CH11-FIL-008

Type: Fill in the Blanks

Difficulty: Easy

Bloom Level: Remember

Topic: Tadpole Breathing Organ

Marks: 1

**Question:** A tadpole breathes with \_\_\_\_\_.

### Question 9

Question ID: CH11-FIL-009

Type: Fill in the Blanks

Difficulty: Easy

Bloom Level: Remember

Topic: Intermediate Stage Name

Marks: 1

**Question:** A tadpole grows into a \_\_\_\_\_.

### Question 10

Question ID: CH11-FIL-010

Type: Fill in the Blanks

Difficulty: Easy

Bloom Level: Remember

Topic: Froglet Movement Adaptation

Marks: 1

**Question:** A froglet has developed legs for \_\_\_\_\_.

### Question 11

Question ID: CH11-FIL-011

Type: Fill in the Blanks

Difficulty: Easy

Bloom Level: Remember

Topic: Froglet Breathing Organ

Marks: 1

**Question:** A froglet has \_\_\_\_\_ which help it breathe air.

### Question 12

Question ID: CH11-FIL-012

Type: Fill in the Blanks

Difficulty: Easy

Bloom Level: Remember

Topic: Final Stage Name

Marks: 1

**Question:** A froglet grows into an adult \_\_\_\_\_.

### Question 13

Question ID: CH11-FIL-013

Type: Fill in the Blanks

Difficulty: Easy

Bloom Level: Remember

Topic: Adult Frog Tail Feature

Marks: 1

**Question:** An adult frog has no \_\_\_\_\_.

### Question 14

Question ID: CH11-FIL-014

Type: Fill in the Blanks

Difficulty: Medium

Bloom Level: Understand

Topic: Vocabulary Meaning - Metamorphosis

Marks: 1

**Question:** The process of changing from one form to another is called \_\_\_\_\_.

### Question 15

Question ID: CH11-FIL-015

Type: Fill in the Blanks

Difficulty: Medium

Bloom Level: Understand

Topic: Vocabulary Meaning - Hatch

Marks: 1

**Question:** Coming out of an egg is called \_\_\_\_\_.

### Question 16

Question ID: CH11-FIL-016

Type: Fill in the Blanks

Difficulty: Easy

Bloom Level: Remember

Topic: Cycle Restart Trigger

Marks: 1

**Question:** A new life cycle begins when a female frog lays \_\_\_\_\_.

### Question 17

Question ID: CH11-FIL-017

Type: Fill in the Blanks

Difficulty: Medium

Bloom Level: Understand

Topic: Spawning Term

Marks: 1

**Question:** Female frogs laying eggs in water is also called \_\_\_\_\_.

### Question 18

Question ID: CH11-FIL-018

Type: Fill in the Blanks

Difficulty: Medium

Bloom Level: Apply

Topic: Organ Development Order

Marks: 1

**Question:** A tadpole breathes with gills first, and later develops \_\_\_\_\_ to breathe air.

### Question 19

Question ID: CH11-FIL-019

Type: Fill in the Blanks

Difficulty: Medium

Bloom Level: Apply

Topic: Leg Development Order

Marks: 1

**Question:** Later on, the tadpole develops \_\_\_\_\_ as it transforms.

### Question 20

Question ID: CH11-FIL-020

Type: Fill in the Blanks

Difficulty: Medium

Bloom Level: Understand

Topic: Cloze Item

Marks: 1

**Question:** From the eggs, tadpoles hatch after one to \_\_\_\_\_ weeks.

### Question 21

Question ID: CH11-FIL-021

Type: Fill in the Blanks

Difficulty: Medium

Bloom Level: Apply

Topic: Grammar - Verb Form

Marks: 1

**Question:** A frog \_\_\_\_\_ through four stages of life.

### Question 22

Question ID: CH11-FIL-022

Type: Fill in the Blanks

Difficulty: Easy

Bloom Level: Remember

Topic: Summary

Marks: 1

**Question:** The four stages of a frog's life are egg, tadpole, froglet, and adult \_\_\_\_\_.

### Question 23

Question ID: CH11-FIL-023

Type: Fill in the Blanks

Difficulty: Hard

Bloom Level: Analyze

Topic: Biological Classification

Marks: 1

**Question:** Animals like frogs that can live both in water and on land are called \_\_\_\_\_.

### Question 24

Question ID: CH11-FIL-024

Type: Fill in the Blanks

Difficulty: Easy

Bloom Level: Remember

Topic: Summary

Marks: 1

**Question:** A froglet has developed legs for \_\_\_\_\_ on land.

### Question 25

Question ID: CH11-FIL-025

Type: Fill in the Blanks

Difficulty: Easy

Bloom Level: Remember

Topic: Summary

Marks: 1

**Question:** Unlike the tadpole, an adult frog has no \_\_\_\_\_ attached to its body.

## Answer Key

| **\*\*CH11-FIL-001\*\*** | **\*\*four / 4\*\*** | Four stages of life. |  
| **\*\*CH11-FIL-002\*\*** | **\*\*metamorphosis\*\*** | Metamorphosis. |  
| **\*\*CH11-FIL-003\*\*** | **\*\*Female\*\*** | Female frogs lay eggs. |  
| **\*\*CH11-FIL-004\*\*** | **\*\*three / 3\*\*** | 1 to 3 weeks. |  
| **\*\*CH11-FIL-005\*\*** | **\*\*Tadpoles\*\*** | Tadpoles hatch. |  
| **\*\*CH11-FIL-006\*\*** | **\*\*legs\*\*** | Tail but no legs. |  
| **\*\*CH11-FIL-007\*\*** | **\*\*water\*\*** | Lives in water. |  
| **\*\*CH11-FIL-008\*\*** | **\*\*gills\*\*** | Breathes with gills. |  
| **\*\*CH11-FIL-009\*\*** | **\*\*froglet\*\*** | Grows into a froglet. |  
| **\*\*CH11-FIL-010\*\*** | **\*\*hopping\*\*** | Legs for hopping. |  
| **\*\*CH11-FIL-011\*\*** | **\*\*lungs\*\*** | Has lungs to breathe. |  
| **\*\*CH11-FIL-012\*\*** | **\*\*frog\*\*** | Adult frog. |  
| **\*\*CH11-FIL-013\*\*** | **\*\*tail\*\*** | Has no tail. |  
| **\*\*CH11-FIL-014\*\*** | **\*\*metamorphosis\*\*** | Word meaning: Metamorphosis. |  
| **\*\*CH11-FIL-015\*\*** | **\*\*hatching / hatch\*\*** | Word meaning: Hatch. |  
| **\*\*CH11-FIL-016\*\*** | **\*\*eggs\*\*** | Female frog lays eggs. |  
| **\*\*CH11-FIL-017\*\*** | **\*\*spawning\*\*** | Egg laying in water = spawning. |  
| **\*\*CH11-FIL-018\*\*** | **\*\*lungs\*\*** | Develops lungs later. |  
| **\*\*CH11-FIL-019\*\*** | **\*\*legs\*\*** | Develops legs. |  
| **\*\*CH11-FIL-020\*\*** | **\*\*three / 3\*\*** | One to three weeks. |  
| **\*\*CH11-FIL-021\*\*** | **\*\*goes\*\*** | A frog goes through. |  
| **\*\*CH11-FIL-022\*\*** | **\*\*frog\*\*** | Adult frog. |  
| **\*\*CH11-FIL-023\*\*** | **\*\*amphibians\*\*** | Living on land and water = amphibians. |  
| **\*\*CH11-FIL-024\*\*** | **\*\*hopping\*\*** | Legs for hopping. |  
| **\*\*CH11-FIL-025\*\*** | **\*\*tail\*\*** | Adult frog has no tail. |

### 3. True or False Statements

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#### Question 1

Question ID: CH11-TF-001

Type: True or False

Difficulty: Easy

Bloom Level: Remember

Topic: Stage Count Check

Marks: 1

**Question:** A frog's life cycle has five stages.

- (A) True
- (B) False

#### Question 2

Question ID: CH11-TF-002

Type: True or False

Difficulty: Easy

Bloom Level: Remember

Topic: Stage Count Check

Marks: 1

**Question:** A frog goes through four stages of life.

- (A) True
- (B) False

#### Question 3

Question ID: CH11-TF-003

Type: True or False

Difficulty: Easy

Bloom Level: Remember

Topic: Tadpole Body Parts Check

Marks: 1

**Question:** Tadpoles have tails but no legs.

- (A) True
- (B) False

#### Question 4

Question ID: CH11-TF-004

Type: True or False

Difficulty: Easy

Bloom Level: Remember

Topic: Tadpole Habitat Check

Marks: 1

**Question:** Tadpoles live on land.



- (A) True
- (B) False

### Question 5

Question ID: CH11-TF-005

Type: True or False

Difficulty: Easy

Bloom Level: Remember

Topic: Tadpole Habitat Check

Marks: 1

**Question:** Tadpoles live in water and breathe with gills.

- (A) True
- (B) False

### Question 6

Question ID: CH11-TF-006

Type: True or False

Difficulty: Medium

Bloom Level: Understand

Topic: Froglet Organ Check

Marks: 1

**Question:** A froglet breathes through gills.

- (A) True
- (B) False

### Question 7

Question ID: CH11-TF-007

Type: True or False

Difficulty: Medium

Bloom Level: Understand

Topic: Froglet Organ Check

Marks: 1

**Question:** A froglet has developed lungs which help it breathe air.

- (A) True
- (B) False

### Question 8

Question ID: CH11-TF-008

Type: True or False

Difficulty: Easy

Bloom Level: Remember

Topic: Adult Frog Tail Check

Marks: 1

**Question:** Adult frogs have tails.

- (A) True
- (B) False

### Question 9

Question ID: CH11-TF-009

Type: True or False

Difficulty: Easy

Bloom Level: Remember

Topic: Adult Frog Tail Check

Marks: 1

**Question:** An adult frog has no tail.

- (A) True
- (B) False

### Question 10

Question ID: CH11-TF-010

Type: True or False

Difficulty: Easy

Bloom Level: Remember

Topic: Metamorphosis Definition Check

Marks: 1

**Question:** Metamorphosis means changing form.

- (A) True
- (B) False

### Question 11

Question ID: CH11-TF-011

Type: True or False

Difficulty: Medium

Bloom Level: Understand

Topic: Hatching Duration Check

Marks: 1

**Question:** It takes 4 to 6 weeks for frog eggs to hatch.

- (A) True
- (B) False

### Question 12

Question ID: CH11-TF-012

Type: True or False

Difficulty: Medium

Bloom Level: Understand

Topic: Hatching Duration Check

Marks: 1

**Question:** It takes from one to three weeks for frog eggs to hatch.

- (A) True
- (B) False

### Question 13

Question ID: CH11-TF-013

Type: True or False

Difficulty: Easy

Bloom Level: Remember

Topic: Spawning Gender Check

Marks: 1

**Question:** Female frogs lay eggs.

- (A) True
- (B) False

### Question 14

Question ID: CH11-TF-014

Type: True or False

Difficulty: Easy

Bloom Level: Remember

Topic: Textbook Exercise C1 Check

Marks: 1

**Question:** Statement C1 in textbook exercise that a frog's life cycle has five stages is False.

- (A) True
- (B) False

### Question 15

Question ID: CH11-TF-015

Type: True or False

Difficulty: Easy

Bloom Level: Remember

Topic: Textbook Exercise C2 Check

Marks: 1

**Question:** Statement C2 in textbook exercise that tadpoles have tails but no legs is True.

- (A) True
- (B) False

### Question 16

Question ID: CH11-TF-016

Type: True or False

Difficulty: Easy

Bloom Level: Remember

Topic: Textbook Exercise C3 Check

Marks: 1

**Question:** Statement C3 in textbook exercise that a froglet breathes through gills is False.

- (A) True
- (B) False

### Question 17

Question ID: CH11-TF-017

Type: True or False

Difficulty: Easy

Bloom Level: Remember

Topic: Textbook Exercise C4 Check

Marks: 1

**Question:** Statement C4 in textbook exercise that adult frogs have tails is False.

- (A) True
- (B) False

### Question 18

Question ID: CH11-TF-018

Type: True or False

Difficulty: Easy

Bloom Level: Remember

Topic: Textbook Exercise C5 Check

Marks: 1

**Question:** Statement C5 in textbook exercise that metamorphosis means changing form is True.

- (A) True
- (B) False

### Question 19

Question ID: CH11-TF-019

Type: True or False

Difficulty: Easy

Bloom Level: Remember

Topic: Textbook Exercise C6 Check

Marks: 1

**Question:** Statement C6 in textbook exercise that tadpoles live on land is False.

- (A) True
- (B) False

### Question 20

Question ID: CH11-TF-020

Type: True or False

Difficulty: Hard

Bloom Level: Evaluate

Topic: Amphibian Lifestyle

Marks: 1

**Question:** A frog can survive both in water during early stages and on land as an adult.

- (A) True
- (B) False

### Question 21

Question ID: CH11-TF-021

Type: True or False

Difficulty: Medium

Bloom Level: Understand

Topic: Froglet Hopping Adaptation

Marks: 1

**Question:** The legs developed by a froglet are used for hopping on land.

- (A) True
- (B) False

### Question 22

Question ID: CH11-TF-022

Type: True or False

Difficulty: Medium

Bloom Level: Understand

Topic: Spawning Term Check

Marks: 1

**Question:** Egg laying by female frogs is also called spawning.

- (A) True
- (B) False

### Question 23

Question ID: CH11-TF-023

Type: True or False

Difficulty: Hard

Bloom Level: Evaluate

Topic: Tail Loss Process

Marks: 1

**Question:** As a froglet transforms into an adult frog, its tail disappears completely.

- (A) True
- (B) False

### Question 24

Question ID: CH11-TF-024

Type: True or False

Difficulty: Easy

Bloom Level: Remember

Topic: Cycle Continuation Check

Marks: 1

**Question:** The lifecycle ends forever when an adult frog dies without laying eggs.

- (A) True
- (B) False

### Question 25

Question ID: CH11-TF-025

Type: True or False

Difficulty: Hard

Bloom Level: Evaluate

Topic: Metamorphosis Necessity

Marks: 1

**Question:** Metamorphosis allows a frog to adapt from an aquatic lifestyle (water breathing with gills) to a terrestrial lifestyle (land breathing with lungs).

- (A) True
- (B) False

## Answer Key

| \*\*CH11-TF-001\*\* | \*\*(B) False\*\* | It has 4 stages, not 5. |

| \*\*CH11-TF-002\*\* | \*\*(A) True\*\* | Four stages of life. |

| \*\*CH11-TF-003\*\* | \*\*(A) True\*\* | Tadpole has tail but no legs. |

| \*\*CH11-TF-004\*\* | \*\*(B) False\*\* | Tadpoles live in water, not on land. |

| \*\*CH11-TF-005\*\* | \*\*(A) True\*\* | Water dweller breathing with gills. |

| \*\*CH11-TF-006\*\* | \*\*(B) False\*\* | Froglet breathes through lungs, not gills. |

| \*\*CH11-TF-007\*\* | \*\*(A) True\*\* | Froglet develops lungs. |

| \*\*CH11-TF-008\*\* | \*\*(B) False\*\* | Adult frogs have no tail. |

| \*\*CH11-TF-009\*\* | \*\*(A) True\*\* | Adult frog has no tail. |

| \*\*CH11-TF-010\*\* | \*\*(A) True\*\* | Metamorphosis = changing form. |

| \*\*CH11-TF-011\*\* | \*\*(B) False\*\* | Takes 1 to 3 weeks, not 4 to 6. |

| \*\*CH11-TF-012\*\* | \*\*(A) True\*\* | Takes 1 to 3 weeks to hatch. |

| \*\*CH11-TF-013\*\* | \*\*(A) True\*\* | Female frogs lay eggs. |

| \*\*CH11-TF-014\*\* | \*\*(A) True\*\* | Statement C1 is False (4 stages). |

| \*\*CH11-TF-015\*\* | \*\*(A) True\*\* | Statement C2 is True. |

| \*\*CH11-TF-016\*\* | \*\*(A) True\*\* | Statement C3 is False (breathes lungs). |

| \*\*CH11-TF-017\*\* | \*\*(A) True\*\* | Statement C4 is False (no tail). |

| \*\*CH11-TF-018\*\* | \*\*(A) True\*\* | Statement C5 is True. |

| \*\*CH11-TF-019\*\* | \*\*(A) True\*\* | Statement C6 is False (lives in water). |

| \*\*CH11-TF-020\*\* | \*\*(A) True\*\* | Amphibian lifestyle. |

| \*\*CH11-TF-021\*\* | \*\*(A) True\*\* | Legs developed for hopping. |

| \*\*CH11-TF-022\*\* | \*\*(A) True\*\* | Egg laying is called spawning. |

| \*\*CH11-TF-023\*\* | \*\*(A) True\*\* | Tail resorbs/disappears into adult frog. |

| **CH11-TF-024** | **(B) False** | Cycle continues as new eggs are laid by female frogs. |

| **CH11-TF-025** | **(A) True** | Essential adaptation from water to land. |

## 4. Short Answer Questions

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### Question 1

Question ID: CH11-SA-001

Type: Short Answer

Difficulty: Easy

Bloom Level: Remember

Topic: Stage Count & Names

Marks: 2

**Question:** How many stages are there in a frog's life cycle? Name all four stages in order.

### Question 2

Question ID: CH11-SA-002

Type: Short Answer

Difficulty: Easy

Bloom Level: Remember

Topic: Definition of Metamorphosis

Marks: 2

**Question:** What is **metamorphosis**, and why is it essential for a frog's development?

### Question 3

Question ID: CH11-SA-003

Type: Short Answer

Difficulty: Easy

Bloom Level: Remember

Topic: First Stage (Egg & Spawning)

Marks: 2

**Question:** Describe the first stage of a frog's life cycle. What is egg laying by female frogs called?

### Question 4

Question ID: CH11-SA-004

Type: Short Answer

Difficulty: Easy

Bloom Level: Remember

Topic: Egg Hatching Duration

Marks: 2

**Question:** How long does it take for frog eggs to hatch, and what creature emerges from them?

### Question 5

Question ID: CH11-SA-005

Type: Short Answer

Difficulty: Easy

Bloom Level: Remember

Topic: Tadpole Features & Habitat

Marks: 2



**Question:** Where do tadpoles live, and how do they breathe in their environment?

### Question 6

Question ID: CH11-SA-006

Type: Short Answer

Difficulty: Medium

Bloom Level: Understand

Topic: Changes from Tadpole to Froglet

Marks: 2

**Question:** What physical changes occur when a tadpole transforms into a froglet?

### Question 7

Question ID: CH11-SA-007

Type: Short Answer

Difficulty: Medium

Bloom Level: Understand

Topic: Difference between Froglet & Adult Frog

Marks: 2

**Question:** What is the main physical difference between a froglet and an adult frog?

### Question 8

Question ID: CH11-SA-008

Type: Short Answer

Difficulty: Medium

Bloom Level: Understand

Topic: Lifecycle Restart Trigger

Marks: 2

**Question:** When does a new life cycle of a frog begin?

### Question 9

Question ID: CH11-SA-009

Type: Short Answer

Difficulty: Medium

Bloom Level: Understand

Topic: Vocabulary Meaning - Metamorphosis & Hatch

Marks: 2

**Question:** Define the textbook terms **'metamorphosis'** and **'hatch'**.

### Question 10

Question ID: CH11-SA-010

Type: Short Answer

Difficulty: Hard

Bloom Level: Evaluate

Topic: Importance of Metamorphosis

Marks: 2

**Question:** Why do you think metamorphosis is important for a frog's survival?

### Question 11

Question ID: CH11-SA-011

Type: Short Answer

Difficulty: Medium

Bloom Level: Understand

Topic: Organ Shift - Gills to Lungs

Marks: 2

**Question:** Why does a frog switch from breathing with gills to breathing with lungs as it matures?

### Question 12

Question ID: CH11-SA-012

Type: Short Answer

Difficulty: Medium

Bloom Level: Apply

Topic: Grammar - Voice Transformation

Marks: 2

**Question:** Change the sentence into Passive Voice:

\*"Female frogs lay eggs in water."\*

### Question 13

Question ID: CH11-SA-013

Type: Short Answer

Difficulty: Medium

Bloom Level: Apply

Topic: Grammar - Tense Transformation

Marks: 2

**Question:** Rewrite the sentence in Simple Past Tense:

\*"A tadpole lives in water and breathes with gills."\*

### Question 14

Question ID: CH11-SA-014

Type: Short Answer

Difficulty: Medium

Bloom Level: Apply

Topic: Grammar - Nouns Identification

Marks: 2

**Question:** Identify two Common Nouns and two Action Verbs in:

\*"A froglet grows legs for hopping on land."\*

### Question 15

Question ID: CH11-SA-015

Type: Short Answer

Difficulty: Hard

Bloom Level: Analyze

Topic: Amphibian Meaning

Marks: 2

**Question:** Why are frogs classified as **\*\*amphibians\*\***?

### Question 16

Question ID: CH11-SA-016

Type: Short Answer

Difficulty: Hard

Bloom Level: Evaluate

Topic: Role of Water in Life Cycle

Marks: 2

**Question:** Why is clean water necessary for female frogs and hatching tadpoles?

### Question 17

Question ID: CH11-SA-017

Type: Short Answer

Difficulty: Medium

Bloom Level: Apply

Topic: Grammar - Antonyms

Marks: 2

**Question:** Write antonyms for the following text words:

(a) **\*\*Female\*\***

(b) **\*\*Adult\*\***

### Question 18

Question ID: CH11-SA-018

Type: Short Answer

Difficulty: Medium

Bloom Level: Apply

Topic: Grammar - Punctuation

Marks: 2

**Question:** Punctuate correctly:

**\*a tadpole grows into a froglet which develops lungs and legs\***

### Question 19

Question ID: CH11-SA-019

Type: Short Answer

Difficulty: Hard

Bloom Level: Analyze

Topic: Function of Hopping Legs

Marks: 2

**Question:** How do strong hind legs help an adult frog survive on land?

### Question 20

Question ID: CH11-SA-020

Type: Short Answer

*Difficulty: Medium*

*Bloom Level: Understand*

*Topic: Tail Resorption*

*Marks: 2*

**Question:** What happens to the tadpole's tail as it becomes an adult frog?

### Question 21

*Question ID: CH11-SA-021*

*Type: Short Answer*

*Difficulty: Hard*

*Bloom Level: Evaluate*

*Topic: Food Habit Shift*

*Marks: 2*

**Question:** How does a tadpole's diet (eating algae in water) differ from an adult frog's diet (catching insects on land)?

### Question 22

*Question ID: CH11-SA-022*

*Type: Short Answer*

*Difficulty: Medium*

*Bloom Level: Apply*

*Topic: Singular to Plural Conversion*

*Marks: 2*

**Question:** Change the following singular terms into plurals:

(a) **Egg**  $\rightarrow$  \_\_\_\_\_

(b) **Froglet**  $\rightarrow$  \_\_\_\_\_

### Question 23

*Question ID: CH11-SA-023*

*Type: Short Answer*

*Difficulty: Medium*

*Bloom Level: Apply*

*Topic: Preposition Fill*

*Marks: 2*

**Question:** Fill in suitable prepositions:

**"Tadpoles hatch \_\_\_\_\_ eggs and live \_\_\_\_\_ water."**

### Question 24

*Question ID: CH11-SA-024*

*Type: Short Answer*

*Difficulty: Medium*

*Bloom Level: Apply*

*Topic: Degrees of Comparison*

*Marks: 2*

**Question:** Complete the degrees of comparison:

(a) **Young**, \_\_\_\_\_, **Youngest**

(b) **Long**, \_\_\_\_\_, **Longest**

## Question 25

Question ID: CH11-SA-025

Type: Short Answer

Difficulty: Easy

Bloom Level: Remember

Topic: Chapter Summary

Marks: 2

**Question:** Write a 2-sentence summary of the scientific chapter **"Lifecycle of a Frog"**.

## Answer Key & Model Answers

| **CH11-SA-001** | There are 4 stages in a frog's life cycle: 1. Egg, 2. Tadpole, 3. Froglet, and 4. Adult Frog. |

| **CH11-SA-002** | Metamorphosis is the biological process of changing form from one stage to another as an animal grows. |

| **CH11-SA-003** | The first stage is the Egg. Female frogs lay jelly-like eggs in water, a process called **\*spawning\***. |

| **CH11-SA-004** | It takes 1 to 3 weeks for eggs to hatch. Swimming tadpoles hatch out from the eggs. |

| **CH11-SA-005** | Tadpoles live in water. They have tails but no legs, and they breathe through gills like fish. |

| **CH11-SA-006** | A tadpole grows legs for hopping, develops lungs for breathing air, and its tail begins to shrink. |

| **CH11-SA-007** | A froglet still has a small shrinking tail, whereas an adult frog has completely lost its tail. |

| **CH11-SA-008** | A new life cycle begins when an adult female frog lays eggs in water. |

| **CH11-SA-009** | 'Metamorphosis' means changing form as an organism grows; 'hatch' means coming out of an egg. |

| **CH11-SA-010** | Metamorphosis allows the frog to adapt from an underwater plant-eating tadpole to a land-hopping insect-eating adult frog. |

| **CH11-SA-011** | Gills absorb oxygen from water, but lungs are needed to breathe atmospheric oxygen once the frog moves onto land. |

| **CH11-SA-012** | **\*Eggs are laid in water by female frogs.\*** |

| **CH11-SA-013** | **\*A tadpole lived in water and breathed with gills.\*** |

| **CH11-SA-014** | **\*\*Common Nouns\*\*:** froglet, legs, land.

**\*\*Action Verbs\*\*:** grows, hopping. |

| **CH11-SA-015** | Frogs are amphibians because they spend their early life in water (tadpole) and adult life on both land and water. |

| **CH11-SA-016** | Eggs need moist water to prevent drying out, and tadpoles need clean water to breathe through gills. |

| **CH11-SA-017** | (a) **\*\*Female\*\*:** Male

(b) **\*\*Adult\*\*:** Young / Child |

| **CH11-SA-018** | **\*A tadpole grows into a froglet, which develops lungs and legs.\*** |

| **CH11-SA-019** | Strong hind legs allow adult frogs to hop fast on land to escape predators and catch moving insects. |

| **CH11-SA-020** | The tail is gradually absorbed into the body as a source of nourishment during the froglet stage until it disappears. |

| **CH11-SA-021** | Tadpoles are plant-eaters (herbivores) eating algae in water, while adult frogs are meat-eaters (carnivores) catching insects. |

| **CH11-SA-022** | (a) Egg → **Eggs**

(b) Froglet → **Froglings** |

| **CH11-SA-023** | **"Tadpoles hatch from eggs and live in water."** |

| **CH11-SA-024** | (a) Young, **Younger**, Youngest

(b) Long, **Longer**, Longest |

| **CH11-SA-025** | A frog undergoes metamorphosis across four life stages: Egg, Tadpole, Froglet, and Adult Frog. Starting in water with gills, it develops legs and lungs to live on land, restarting the cycle when eggs are laid. |

## 5. Long Answer & HOTS Questions

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### Question 1

Question ID: CH11-LA-001

Type: Long Answer & HOTS

Difficulty: Medium

Bloom Level: Understand

Topic: Comprehensive Lifecycle Narrative

Marks: 5

**Question:** Describe the complete 4-stage lifecycle of a frog from egg to adult frog. Explain the physical changes at each stage.

### Question 2

Question ID: CH11-LA-002

Type: Long Answer & HOTS

Difficulty: Hard

Bloom Level: Evaluate

Topic: Metamorphosis Significance

Marks: 5

**Question:** What is **metamorphosis**? Evaluate why metamorphosis is a crucial biological adaptation for frogs moving from aquatic to terrestrial life.

### Question 3

Question ID: CH11-LA-003

Type: Long Answer & HOTS

Difficulty: Hard

Bloom Level: Analyze

Topic: Respiratory Shift (Gills to Lungs)

Marks: 5

**Question:** Trace the respiratory transformation in a frog's life cycle. Compare how a tadpole breathes underwater with how a froglet and adult frog breathe air on land.

### Question 4

Question ID: CH11-LA-004

Type: Long Answer & HOTS

Difficulty: Hard

Bloom Level: Analyze

Topic: Anatomical Changes (Tail & Legs)

Marks: 5

**Question:** Compare the anatomical features of a tadpole (tail, no legs) with an adult frog (no tail, strong legs). Why do these changes occur during metamorphosis?

### Question 5

Question ID: CH11-LA-005

Type: Long Answer & HOTS

*Difficulty: Medium*

*Bloom Level: Understand*

*Topic: Role of Water in Spawning & Hatching*

*Marks: 5*

**Question:** Explain why female frogs lay their eggs (\*spawning\*) in water. What environment do eggs and young tadpoles require to survive and hatch within 1 to 3 weeks?

### Question 6

*Question ID: CH11-LA-006*

*Type: Long Answer & HOTS*

*Difficulty: Hard*

*Bloom Level: Create*

*Topic: First-Person Tadpole Story*

*Marks: 5*

**Question:** Imagine you are a little tadpole swimming in a pond. Write a 6-sentence story describing your daily life, breathing through gills, and growing legs as you turn into a froglet.

### Question 7

*Question ID: CH11-LA-007*

*Type: Long Answer & HOTS*

*Difficulty: Hard*

*Bloom Level: Evaluate*

*Topic: Amphibians & Ecosystem Balance*

*Marks: 5*

**Question:** Frogs are amphibians that control insect populations. Evaluate the ecological importance of protecting freshwater ponds where frogs reproduce.

### Question 8

*Question ID: CH11-LA-008*

*Type: Long Answer & HOTS*

*Difficulty: Medium*

*Bloom Level: Understand*

*Topic: Vocabulary & Theme Analysis*

*Marks: 5*

**Question:** Explain the key scientific terms: **metamorphosis**, **hatch**, **spawning**, **tadpole**, and **froglet**. How does each term relate to Chapter 11?

### Question 9

*Question ID: CH11-LA-009*

*Type: Long Answer & HOTS*

*Difficulty: Hard*

*Bloom Level: Create*

*Topic: Imaginary Dialogue*

*Marks: 5*

**Question:** Write a fictional 4-exchange dialogue between a fish and a newly transformed adult frog discussing the differences between living only in water versus living on land.



### Question 10

Question ID: CH11-LA-010

Type: Long Answer & HOTS

Difficulty: Hard

Bloom Level: Evaluate

Topic: Comparison - Butterfly vs Frog Metamorphosis

Marks: 5

**Question:** Compare the metamorphosis of a butterfly (Egg  $\rightarrow$  Caterpillar  $\rightarrow$  Pupa  $\rightarrow$  Butterfly) with the metamorphosis of a frog (Egg  $\rightarrow$  Tadpole  $\rightarrow$  Froglet  $\rightarrow$  Frog).

### Question 11

Question ID: CH11-LA-011

Type: Long Answer & HOTS

Difficulty: Medium

Bloom Level: Apply

Topic: Science Fair Poster Proposal

Marks: 5

**Question:** Design a 5-point proposal for a Class IV Science Fair poster titled *"The Magic of Metamorphosis: How a Frog Grows"*.

### Question 12

Question ID: CH11-LA-012

Type: Long Answer & HOTS

Difficulty: Hard

Bloom Level: Create

Topic: Formal Scientific Report

Marks: 5

**Question:** Write a formal 6-sentence scientific summary explaining the life cycle and biological changes of *"Chapter 11: Lifecycle of a Frog"*.

### Question 13

Question ID: CH11-LA-013

Type: Long Answer & HOTS

Difficulty: Medium

Bloom Level: Understand

Topic: Text Structure Analysis

Marks: 5

**Question:** How does the author structure Chapter 11? Trace the narrative from definition of metamorphosis  $\rightarrow$  Stage 1 (Egg)  $\rightarrow$  Stage 2 (Tadpole)  $\rightarrow$  Stage 3 (Froglet)  $\rightarrow$  Stage 4 (Adult Frog)  $\rightarrow$  Cycle continuation.

### Question 14

Question ID: CH11-LA-014

Type: Long Answer & HOTS

Difficulty: Hard

*Bloom Level: Evaluate*

*Topic: Environmental Hazards to Tadpoles*

*Marks: 5*

**Question:** Discuss the hazards that frog eggs and young tadpoles face in ponds (such as water pollution, predators, and drying ponds). How can humans protect them?

### **Question 15**

*Question ID: CH11-LA-015*

*Type: Long Answer & HOTS*

*Difficulty: Hard*

*Bloom Level: Analyze*

*Topic: Locomotion Transformation*

*Marks: 5*

**Question:** Analyze how locomotion changes across a frog's life cycle—from swimming with a tail as a tadpole to hopping on land with long hind legs as an adult frog.

### **Question 16**

*Question ID: CH11-LA-016*

*Type: Long Answer & HOTS*

*Difficulty: Medium*

*Bloom Level: Apply*

*Topic: Classroom Pond Experiment Safety*

*Marks: 5*

**Question:** If a science teacher brings a glass jar of pond water with tadpoles to class, list 5 safety and care rules for observing the tadpoles gently.

### **Question 17**

*Question ID: CH11-LA-017*

*Type: Long Answer & HOTS*

*Difficulty: Hard*

*Bloom Level: Evaluate*

*Topic: Why Frogs Need Both Land and Water*

*Marks: 5*

**Question:** Explain why a frog cannot complete its life cycle if either water or land is missing from its habitat.

### **Question 18**

*Question ID: CH11-LA-018*

*Type: Long Answer & HOTS*

*Difficulty: Hard*

*Bloom Level: Create*

*Topic: Descriptive Observation Journal*

*Marks: 5*

**Question:** Imagine you maintained an observation journal over 6 weeks watching frog eggs hatch into tadpoles and grow into froglets. Write a 5-entry journal summary.

### Question 19

Question ID: CH11-LA-019

Type: Long Answer & HOTS

Difficulty: Medium

Bloom Level: Understand

Topic: Cause and Effect Analysis

Marks: 5

**Question:** Map 3 key cause-and-effect biological transformations detailed in Chapter 11.

### Question 20

Question ID: CH11-LA-020

Type: Long Answer & HOTS

Difficulty: Hard

Bloom Level: Evaluate

Topic: Tail Absorption Biology

Marks: 5

**Question:** Explain what happens to the tadpole's tail during the froglet stage. How does absorbing the tail benefit the growing froglet?

### Question 21

Question ID: CH11-LA-021

Type: Long Answer & HOTS

Difficulty: Medium

Bloom Level: Apply

Topic: School Assembly Quiz Design

Marks: 5

**Question:** Design 5 trivia quiz questions based on Chapter 11 for a school morning assembly nature and science quiz.

### Question 22

Question ID: CH11-LA-022

Type: Long Answer & HOTS

Difficulty: Hard

Bloom Level: Analyze

Topic: Continuous Cycle Concept

Marks: 5

**Question:** Analyze the phrase: "A new life cycle begins when a female frog lays eggs." Why is the concept of a "cycle" fundamental to all living nature?

### Question 23

Question ID: CH11-LA-023

Type: Long Answer & HOTS

Difficulty: Hard

Bloom Level: Create

Topic: Conservation Slogan Campaign

Marks: 5

**Question:** Create a 5-point slogan campaign encouraging young students to protect pond ecosystems and save frogs.

### Question 24

Question ID: CH11-LA-024

Type: Long Answer & HOTS

Difficulty: Medium

Bloom Level: Understand

Topic: Diet Shift Analysis

Marks: 5

**Question:** Describe how a frog's digestive system and diet adapt from herbivorous tadpoles (eating water plants) to carnivorous adult frogs (catching insects).

### Question 25

Question ID: CH11-LA-025

Type: Long Answer & HOTS

Difficulty: Hard

Bloom Level: Evaluate

Topic: Final Takeaway

Marks: 5

**Question:** Summarize the overall scientific lesson of Chapter 11 for Class IV students regarding growth, adaptation, environmental care, and nature's life cycles.

## Model Answers & Rubrics

| **\*\*CH11-LA-001\*\*** | The lifecycle has four stages: 1. **\*\*Egg\*\***: Female frogs lay jelly-like eggs in water (**\*spawning\***), taking 1-3 weeks to hatch. 2. **\*\*Tadpole\*\***: Hatches with a tail and no legs, living in water and breathing through gills. 3. **\*\*Froglet\*\***: Grows hind and front legs for hopping, develops lungs to breathe air, and has a shrinking tail. 4. **\*\*Adult Frog\*\***: Fully grown with no tail, living on land and in water. The female lays eggs to start a new cycle. | Stage 1 & 2: 2 Marks

Stage 3 & 4: 2 Marks

Cycle continuity: 1 Mark |

| **\*\*CH11-LA-002\*\*** | Metamorphosis is the process of changing form as an organism grows. It is crucial because it allows the frog to transition from an aquatic herbivore (tadpole breathing with gills in water) into a terrestrial carnivore (adult frog breathing with lungs on land), reducing competition for food between young and adults. | Metamorphosis definition: 2 Marks

Biological adaptation value: 3 Marks |

| **\*\*CH11-LA-003\*\*** | In Stage 2 (Tadpole), the organism lives underwater and extracts dissolved oxygen using feather-like **\*\*gills\*\***. In Stage 3 (Froglet) and Stage 4 (Adult Frog), internal **\*\*lungs\*\*** develop, allowing the frog to take in atmospheric oxygen from air when it moves out of water onto land. | Gills in tadpole: 2 Marks

Lungs in froglet/adult: 2 Marks

Transition rationale: 1 Mark |

| **\*\*CH11-LA-004\*\*** | A tadpole has a long, flexible tail for swimming in water and lacks legs. An adult frog has lost its tail completely and developed powerful hind legs for hopping on land and webbed feet for swimming. These anatomical changes match their shifting habitats. | Tadpole features: 2 Marks

Adult features: 2 Marks

Adaptation explanation: 1 Mark |

| **\*\*CH11-LA-005\*\*** | Female frogs lay eggs in water because frog eggs lack hard protective shells and would dry out on land. The moist pond environment provides oxygen and nutrients for the jelly-covered eggs, allowing tadpoles to hatch safely in 1 to 3 weeks. | Water requirement: 3 Marks  
Egg protection & Hatching: 2 Marks |

| **\*\*CH11-LA-006\*\*** | \*Hello, I am a tiny tadpole swimming in a quiet pond! I hatched from a soft egg just two weeks ago. I swim around using my long tail and breathe oxygen underwater through my gills. Everyday I eat small green algae from pond rocks. Lately, I noticed small hind legs growing near my tail! Soon I will develop lungs and hop onto the grassy bank as a froglet.\* | First-person story: 2 Marks  
Biological accuracy: 3 Marks |

| **\*\*CH11-LA-007\*\*** | Adult frogs consume thousands of mosquitoes, flies, and agricultural pests, keeping ecosystems balanced. Tadpoles clean pond water by eating algae. Protecting ponds ensures frogs survive, preventing insect overpopulation and disease spread. | Insect control role: 3 Marks  
Pond protection value: 2 Marks |

| **\*\*CH11-LA-008\*\*** | **\*\*Metamorphosis\*\***: Process of changing form during growth.  
**\*\*Hatch\*\***: Emerging out of an egg shell/jelly.  
**\*\*Spawning\*\***: Female frogs laying egg masses in water.  
**\*\*Tadpole\*\***: Aquatic stage with tail and gills.  
**\*\*Froglet\*\***: Transitional stage with legs, lungs, and small tail. | 5 Word links: 3 Marks  
Contextual accuracy: 2 Marks |

| **\*\*CH11-LA-009\*\*** | **\*\*Fish\*\***: "Hello frog! Why do you leave our cool pond to jump onto the dry grass?"  
**\*\*Adult Frog\*\***: "I have developed lungs and strong legs! Now I can breathe air and hop on land."  
**\*\*Fish\*\***: "Don't you miss breathing underwater with gills like when you were a tadpole?"  
**\*\*Adult Frog\*\***: "My gills transformed into lungs during metamorphosis, but I still visit the pond to lay eggs!" | Dialogue format: 2 Marks  
Scientific contrast: 3 Marks |

| **\*\*CH11-LA-010\*\*** | Both undergo complete metamorphosis. A butterfly goes through Egg → Caterpillar (larva) → Pupa (chrysalis) → Adult Butterfly. A frog goes through Egg → Tadpole → Froglet → Adult Frog. Both change feeding habits and body structures completely. | Butterfly stages: 2 Marks  
Frog stages: 2 Marks  
Comparison: 1 Mark |

| **\*\*CH11-LA-011\*\*** | 1. Draw a large central circular diagram showing the 4 stages.  
2. Add clear labels: Egg (1-3 weeks), Tadpole (gills/tail), Froglet (lungs/legs), Adult Frog (no tail).  
3. Include vocabulary callout boxes defining Metamorphosis and Spawning.  
4. Display photos of clean pond habitats.  
5. Write a summary explaining how lungs replace gills. | 5 Proposal points: 3 Marks  
Educational clarity: 2 Marks |

| **\*\*CH11-LA-012\*\*** | A frog undergoes complete metamorphosis through four distinct life stages. Female frogs lay eggs in water during spawning, which hatch into swimming tadpoles after one to three weeks. Tadpoles live in water, possess tails, and breathe through gills. As the tadpole matures into a froglet, it develops legs for hopping and lungs for breathing atmospheric air. Finally, the froglet loses its tail to become an adult frog. The cycle restarts when female frogs lay eggs in freshwater habitats. | Scientific report: 3 Marks  
Accuracy & Flow: 2 Marks |

| **\*\*CH11-LA-013\*\*** | The text opens with core definition (metamorphosis & 4 stages) → Stage 1 Egg/Spawning (1-3 weeks) → Stage 2 Tadpole (tail, gills, water) → Stage 3 Froglet (legs, lungs, hopping) → Stage 4 Adult Frog (no tail, land) → Life cycle continuation. |  
Text organization: 3 Marks  
Flow: 2 Marks |

| **CH11-LA-014** | Hazards include chemical water pollution, predatory fish eating eggs, and drought drying up ponds before 3 weeks. Humans can help by protecting natural ponds, avoiding dumping trash/chemicals, and creating garden ponds. | Hazards identification: 3 Marks  
Human protection actions: 2 Marks |

| **CH11-LA-015** | Stage 1 (Egg): Stationary → Stage 2 (Tadpole): Lateral tail-wiggling swimming → Stage 3 (Froglet): Combined tail swimming and leg kicking → Stage 4 (Adult Frog): High-power leg hopping on land and webbed-foot kicking in water. | Locomotion stages: 3 Marks  
Adaptation link: 2 Marks |

| **CH11-LA-016** | 1. Use unchlorinated pond water, never tap water.  
2. Keep the jar away from direct hot sunlight.  
3. Feed small amounts of boiled lettuce or algae.  
4. Never touch tadpoles with dry hands.  
5. Release grown froglets back into their original natural pond. | 5 Safety rules: 3 Marks  
Practical care: 2 Marks |

| **CH11-LA-017** | Without water, eggs dry out and tadpoles die because they breathe with gills. Without land, adult frogs cannot find land insects or thrive as amphibians. Both habitats are required to complete all 4 stages. | Water dependency: 2.5 Marks  
Land dependency: 2.5 Marks |

| **CH11-LA-018** | **Week 1**: Found jelly-like egg mass floating in pond.  
**Week 2**: Tiny tadpoles hatched! Swimming with long tails.  
**Week 4**: Tadpoles growing larger, breathing with gills.  
**Week 5**: Small hind legs emerging! Tadpole turning into froglet.  
**Week 6**: Froglet developed lungs and hopped onto floating leaf! | 5 Journal entries: 3 Marks  
Chronology: 2 Marks |

| **CH11-LA-019** | 1. **Cause**: Female lays eggs in water → **Effect**: Tadpoles hatch in 1-3 weeks.  
2. **Cause**: Development of lungs and legs → **Effect**: Froglet can move onto land and breathe air.  
3. **Cause**: Absorption of tail → **Effect**: Froglet transforms into tailless adult frog. | 3  
Cause-Effect pairs: 3 Marks  
Accuracy: 2 Marks |

| **CH11-LA-020** | The tail does not drop off; it is gradually broken down by the body (resorbed). The nutrients and energy stored in the tail are absorbed to nourish the growing legs and internal organs while the froglet stops eating. | Tail absorption mechanism: 3 Marks  
Nutritional benefit: 2 Marks |

| **CH11-LA-021** | 1. **How many stages are there in a frog's life cycle? (Four)**  
2. **What is the process of changing form called? (Metamorphosis)**  
3. **How do tadpoles breathe in water? (With gills)**  
4. **How long does it take for frog eggs to hatch? (1 to 3 weeks)**  
5. **Does an adult frog have a tail? (No)** | 5 Assembly questions: 3 Marks  
Accuracy: 2 Marks |

| **CH11-LA-022** | A cycle has no beginning or end; every adult stage reproduces to generate new eggs, ensuring continuous survival of the species across generations in nature. | Cycle concept analysis: 3 Marks  
Nature connection: 2 Marks |

| **CH11-LA-023** | 1. **Protect our ponds, save frog homes!**  
2. **Clean water means healthy tadpoles!**  
3. **Stop water pollution, let metamorphosis happen!**

4. \*Frogs eat pests, keep nature at rest!\*

5. \*Respect all life stages, from egg to adult frog!\* | 5 Slogans: 3 Marks

Quality: 2 Marks |

| \*\*CH11-LA-024\*\* | Tadpoles have long coiled intestines suited for digesting plant matter (algae). During metamorphosis, as legs and lungs grow, the stomach and intestine shorten to digest protein-rich insects eaten by adult frogs. | Digestive adaptation: 3 Marks

Dietary contrast: 2 Marks |

| \*\*CH11-LA-025\*\* | Chapter 11 teaches that nature is full of incredible transformations. By understanding metamorphosis, students appreciate animal adaptations, the importance of clean habitats, and nature's perpetual life cycles. | Core lesson: 3 Marks

Synthesis: 2 Marks |

## 6. Vocabulary & Word Power

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### Question 1

Question ID: CH11-VOC-001

Type: Vocabulary

Difficulty: Easy

Bloom Level: Remember

Topic: Word Meaning - Metamorphosis

Marks: 1

**Question:** What is the textbook definition of **'metamorphosis'**?

- (A) The process of changing from one form to another
- (B) Coming out of an egg
- (C) Laying eggs in water
- (D) Sleeping underwater

### Question 2

Question ID: CH11-VOC-002

Type: Vocabulary

Difficulty: Easy

Bloom Level: Remember

Topic: Word Meaning - Hatch

Marks: 1

**Question:** What is the textbook definition of **'hatch'**?

- (A) Coming out of egg
- (B) Growing a long tail
- (C) Hopping on land
- (D) Losing legs

### Question 3

Question ID: CH11-VOC-003

Type: Vocabulary

Difficulty: Easy

Bloom Level: Remember

Topic: Word Meaning - Spawning

Marks: 1

**Question:** What does **'spawning'** mean in **'Female frogs lay eggs. This is also called spawning.'**?

- (A) The process of laying a mass of eggs in water
- (B) Swimming fast with a tail
- (C) Breathing with lungs
- (D) Changing skin color



### Question 4

Question ID: CH11-VOC-004

Type: Vocabulary

Difficulty: Easy

Bloom Level: Remember

Topic: Word Meaning - Tadpole

Marks: 1

**Question:** What is a **'tadpole'**?

- (A) The aquatic larval stage of a frog with a tail and gills
- (B) An adult frog with no tail
- (C) A bird egg
- (D) A dry land insect

### Question 5

Question ID: CH11-VOC-005

Type: Vocabulary

Difficulty: Easy

Bloom Level: Remember

Topic: Word Meaning - Froglet

Marks: 1

**Question:** What is a **'froglet'**?

- (A) A young frog that has developed legs and lungs but still retains a small tail
- (B) A tiny egg in water
- (C) A fish without legs
- (D) A fully grown adult frog

### Question 6

Question ID: CH11-VOC-006

Type: Vocabulary

Difficulty: Medium

Bloom Level: Understand

Topic: Synonyms

Marks: 1

**Question:** Choose the synonym for **'change'** in **'as the frog grows, it changes'**.

- (A) Remains
- (B) Transforms / Alters / Develops
- (C) Stops
- (D) Sleeps

### Question 7

Question ID: CH11-VOC-007

Type: Vocabulary

Difficulty: Medium

*Bloom Level: Understand*

*Topic: Synonyms*

*Marks: 1*

**Question:** Choose the synonym for **'adult'** in **'adult frog'**.

- (A) Young
- (B) Fully grown / Mature
- (C) Tiny
- (D) Baby

### Question 8

*Question ID: CH11-VOC-008*

*Type: Vocabulary*

*Difficulty: Medium*

*Bloom Level: Apply*

*Topic: Antonyms*

*Marks: 1*

**Question:** What is the antonym of **'female'** in **'female frog'**?

- (A) Woman
- (B) Male
- (C) Mother
- (D) Girl

### Question 9

*Question ID: CH11-VOC-009*

*Type: Vocabulary*

*Difficulty: Medium*

*Bloom Level: Apply*

*Topic: Antonyms*

*Marks: 1*

**Question:** What is the antonym of **'adult'**?

- (A) Mature
- (B) Juvenile / Young / Baby
- (C) Elderly
- (D) Fully grown

### Question 10

*Question ID: CH11-VOC-010*

*Type: Vocabulary*

*Difficulty: Easy*

*Bloom Level: Remember*

*Topic: Spelling Check*

*Marks: 1*

**Question:** Select the correctly spelled word from Chapter 11.

- (A) Metamorphosis
- (B) Metamorfosis
- (C) Metamorphasis
- (D) Metamorphoses

### Question 11

Question ID: CH11-VOC-011

Type: Vocabulary

Difficulty: Easy

Bloom Level: Remember

Topic: Spelling Check

Marks: 1

**Question:** Select the correct spelling for the young frog stage.

- (A) Froglet
- (B) Froglat
- (C) Froglitt
- (D) Phroglet

### Question 12

Question ID: CH11-VOC-012

Type: Vocabulary

Difficulty: Easy

Bloom Level: Remember

Topic: Spelling Check

Marks: 1

**Question:** Select the correct spelling for the water-breathing organs.

- (A) Gills
- (B) Gils
- (C) Guills
- (D) Jills

### Question 13

Question ID: CH11-VOC-013

Type: Vocabulary

Difficulty: Medium

Bloom Level: Apply

Topic: Word Formation - Diminutive Suffix

Marks: 1

**Question:** What does the suffix **\*\*-let\*\*** mean in **‘froglet’**?

- (A) Large or huge
- (B) Small or young version of something
- (C) Without or lacking

- (D) Full of

### Question 14

Question ID: CH11-VOC-014

Type: Vocabulary

Difficulty: Medium

Bloom Level: Apply

Topic: Word Formation - Suffixes

Marks: 1

**Question:** Add a suffix to **'spawn'** to form a gerund noun meaning egg laying.

- (A) -ing (Spawning)
- (B) -ed
- (C) -ment
- (D) -ness

### Question 15

Question ID: CH11-VOC-015

Type: Vocabulary

Difficulty: Medium

Bloom Level: Understand

Topic: Homophones

Marks: 1

**Question:** Select the correct homophone: "The tadpole grew \_\_\_\_\_ (four/for) legs."

- (A) fore
- (B) four
- (C) for
- (D) fro

### Question 16

Question ID: CH11-VOC-016

Type: Vocabulary

Difficulty: Medium

Bloom Level: Understand

Topic: Homophones

Marks: 1

**Question:** Select the correct word: "The tadpole swims with its long \_\_\_\_\_ (tail/tale)."

- (A) tale
- (B) tail
- (C) teal
- (D) tell

### Question 17

Question ID: CH11-VOC-017

Type: Vocabulary

Difficulty: Hard

Bloom Level: Apply

Topic: Contextual Usage

Marks: 1

**Question:** Which word best completes the sentence?

\*"A tadpole breathes with \_\_\_\_\_ while living underwater."\*

- (A) lungs
- (B) gills
- (C) nose
- (D) wings

### Question 18

Question ID: CH11-VOC-018

Type: Vocabulary

Difficulty: Medium

Bloom Level: Apply

Topic: Greek Origin - Meta & Morph

Marks: 1

**Question:** In the word **'metamorphosis'**, what does **'morph'** mean in ancient Greek?

- (A) Water
- (B) Form / Shape
- (C) Time
- (D) Life

### Question 19

Question ID: CH11-VOC-019

Type: Vocabulary

Difficulty: Hard

Bloom Level: Analyze

Topic: Correct Usage

Marks: 1

**Question:** Which sentence uses the word **'hatch'** correctly?

- (A) The adult frog hatched a story.
- (B) Tadpoles hatch from the frog eggs after one to three weeks.
- (C) We hatched a frog on land.
- (D) Hatch is a stage with four legs.

### Question 20

Question ID: CH11-VOC-020

Type: Vocabulary

Difficulty: Medium

Bloom Level: Apply

Topic: Collocations

Marks: 1

**Question:** Complete the collocation from Page 48:

\*"Four \_\_\_\_\_ of life."\*

- (A) stages
- (B) colors
- (C) legs
- (D) weeks

### Question 21

Question ID: CH11-VOC-021

Type: Vocabulary

Difficulty: Easy

Bloom Level: Remember

Topic: Opposites

Marks: 1

**Question:** What is the opposite of **'begin'** in **'a new life cycle begins'**?

- (A) Start
- (B) End / Finish / Conclude
- (C) Continue
- (D) Open

### Question 22

Question ID: CH11-VOC-022

Type: Vocabulary

Difficulty: Medium

Bloom Level: Understand

Topic: Synonyms

Marks: 1

**Question:** What is a synonym for **'develop'** in **'develops legs for hopping'**?

- (A) Loses
- (B) Grows / Forms / Acquires
- (C) Hides
- (D) Breaks

### Question 23

Question ID: CH11-VOC-023

Type: Vocabulary

Difficulty: Hard

Bloom Level: Analyze

Topic: Term for Water & Land Animals

Marks: 1

**Question:** What scientific term describes animals like frogs that live in both water and land?

- (A) Reptiles
- (B) Amphibians
- (C) Mammals
- (D) Insects

### Question 24

Question ID: CH11-VOC-024

Type: Vocabulary

Difficulty: Medium

Bloom Level: Apply

Topic: Antonyms

Marks: 1

**Question:** What is the antonym of **'aquatic'** (living in water)?

- (A) Marine
- (B) Terrestrial / Land-dwelling
- (C) Oceanic
- (D) Moist

### Question 25

Question ID: CH11-VOC-025

Type: Vocabulary

Difficulty: Medium

Bloom Level: Apply

Topic: Alphabetical Order

Marks: 1

**Question:** Arrange these 4 story words in alphabetical dictionary order:

\*Tadpole, Froglet, Metamorphosis, Egg\*

- (A) Egg, Froglet, Metamorphosis, Tadpole
- (B) Froglet, Egg, Metamorphosis, Tadpole
- (C) Metamorphosis, Egg, Froglet, Tadpole
- (D) Egg, Metamorphosis, Froglet, Tadpole

## Answer Key

| **CH11-VOC-001** | **(A)** | Page 49 word meaning: Metamorphosis = Process of changing from one form to another. |

| **CH11-VOC-002** | **(A)** | Page 49 word meaning: Hatch = Coming out of egg. |

| **CH11-VOC-003** | **(A)** | Spawning = Laying egg mass in water. |

| **CH11-VOC-004** | **(A)** | Tadpole = Aquatic larval stage with tail and gills. |

| **CH11-VOC-005** | **(A)** | Froglet = Young frog with legs, lungs, and small tail. |

| **CH11-VOC-006** | **(B)** | Change means transform, alter, or develop. |

| **CH11-VOC-007** | **(B)** | Adult means fully grown or mature. |

- | **\*\*CH11-VOC-008\*\*** | **\*\* (B) \*\*** | Opposite of female is male. |
- | **\*\*CH11-VOC-009\*\*** | **\*\* (B) \*\*** | Opposite of adult is juvenile or young. |
- | **\*\*CH11-VOC-010\*\*** | **\*\* (A) \*\*** | 'Metamorphosis' is the correct spelling. |
- | **\*\*CH11-VOC-011\*\*** | **\*\* (A) \*\*** | 'Froglet' is the correct spelling. |
- | **\*\*CH11-VOC-012\*\*** | **\*\* (A) \*\*** | 'Gills' is the correct spelling. |
- | **\*\*CH11-VOC-013\*\*** | **\*\* (B) \*\*** | Suffix -let means small/young version (frog + let = young frog). |
- | **\*\*CH11-VOC-014\*\*** | **\*\* (A) \*\*** | Spawn + ing = Spawning. |
- | **\*\*CH11-VOC-015\*\*** | **\*\* (B) \*\*** | 'Four' refers to the number 4. |
- | **\*\*CH11-VOC-016\*\*** | **\*\* (B) \*\*** | 'Tail' is the body part. |
- | **\*\*CH11-VOC-017\*\*** | **\*\* (B) \*\*** | Gills are water-breathing organs. |
- | **\*\*CH11-VOC-018\*\*** | **\*\* (B) \*\*** | Morph in Greek means form or shape. |
- | **\*\*CH11-VOC-019\*\*** | **\*\* (B) \*\*** | Tadpoles hatch coming out of eggs. |
- | **\*\*CH11-VOC-020\*\*** | **\*\* (A) \*\*** | Collocation: "four stages of life". |
- | **\*\*CH11-VOC-021\*\*** | **\*\* (B) \*\*** | Opposite of begin is end or finish. |
- | **\*\*CH11-VOC-022\*\*** | **\*\* (B) \*\*** | Develop means grow, form, or acquire. |
- | **\*\*CH11-VOC-023\*\*** | **\*\* (B) \*\*** | Amphibians live in both water and land. |
- | **\*\*CH11-VOC-024\*\*** | **\*\* (B) \*\*** | Opposite of aquatic is terrestrial (land). |
- | **\*\*CH11-VOC-025\*\*** | **\*\* (A) \*\*** | E (Egg) -> F (Froglet) -> M (Metamorphosis) -> T (Tadpole). |



## 7. Grammar & Language Skills

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### Question 1

Question ID: CH11-GRA-001

Type: Grammar

Difficulty: Easy

Bloom Level: Remember

Topic: Nouns - Common Nouns

Marks: 1

**Question:** Identify the Common Nouns in:

\*"A frog goes through four stages of life."\*

- (A) frog, stages, life
- (B) goes, through
- (C) four, a
- (D) is, was

### Question 2

Question ID: CH11-GRA-002

Type: Grammar

Difficulty: Easy

Bloom Level: Remember

Topic: Nouns - Organs

Marks: 1

**Question:** Identify the Noun referring to a breathing organ in:

\*"A tadpole breathes with gills."\*

- (A) breathes
- (B) gills
- (C) with
- (D) tadpole

### Question 3

Question ID: CH11-GRA-003

Type: Grammar

Difficulty: Medium

Bloom Level: Apply

Topic: Passive to Active Voice

Marks: 1

**Question:** Convert into Active Voice:

\*"Eggs are laid in water by female frogs."\*

- (A) Female frogs lay eggs in water.
- (B) Female frogs laid eggs in water.

- (C) Female frogs will lay eggs in water.
- (D) Female frogs are laying eggs in water.

#### Question 4

Question ID: CH11-GRA-004

Type: Grammar

Difficulty: Medium

Bloom Level: Apply

Topic: Active to Passive Voice

Marks: 1

**Question:** Convert into Passive Voice:

\*"A froglet uses lungs to breathe air."\*

- (A) Lungs are used by a froglet to breathe air.
- (B) Lungs were used by a froglet.
- (C) Lungs will be used by a froglet.
- (D) A froglet was using lungs.

#### Question 5

Question ID: CH11-GRA-005

Type: Grammar

Difficulty: Easy

Bloom Level: Understand

Topic: Adjectives of Number & Feature

Marks: 1

**Question:** Identify the Adjective of Number in:

\*"A frog goes through four stages of life."\*

- (A) frog
- (B) four
- (C) stages
- (D) through

#### Question 6

Question ID: CH11-GRA-006

Type: Grammar

Difficulty: Medium

Bloom Level: Apply

Topic: Degrees of Comparison

Marks: 1

**Question:** Complete the degree of comparison: \*\*young, younger, \_\_\_\_\_\*\*.

- (A) youngest
- (B) most young
- (C) more younger

- (D) youngness

### Question 7

Question ID: CH11-GRA-007

Type: Grammar

Difficulty: Medium

Bloom Level: Apply

Topic: Degrees of Comparison

Marks: 1

**Question:** What is the comparative degree of the adjective **'long'** in **'long tail'**?

- (A) Longer
- (B) Longest
- (C) Most long
- (D) Longness

### Question 8

Question ID: CH11-GRA-008

Type: Grammar

Difficulty: Easy

Bloom Level: Remember

Topic: Verbs - Present Tense

Marks: 1

**Question:** What is the simple present tense of the verb **'hatch'** with a plural subject (**'tadpoles'**)?

- (A) Hatched
- (B) Hatch
- (C) Hatches
- (D) Hatching

### Question 9

Question ID: CH11-GRA-009

Type: Grammar

Difficulty: Easy

Bloom Level: Remember

Topic: Verbs - Past Tense

Marks: 1

**Question:** What is the simple past tense of the verb **'grow'** as in **'a tadpole grew into a froglet'**?

- (A) Growed
- (B) Grew
- (C) Grown
- (D) Growing

### Question 10

Question ID: CH11-GRA-010

Type: Grammar

Difficulty: Medium

Bloom Level: Apply

Topic: Pronouns

Marks: 1

**Question:** Identify the Personal Pronoun in:

\*"As the frog grows, it changes."\*

- (A) frog
- (B) it
- (C) grows
- (D) as

### Question 11

Question ID: CH11-GRA-011

Type: Grammar

Difficulty: Medium

Bloom Level: Apply

Topic: Prepositions

Marks: 1

**Question:** Choose the correct preposition:

\*"A tadpole lives \_\_\_\_\_ water."\*

- (A) under
- (B) in
- (C) between
- (D) over

### Question 12

Question ID: CH11-GRA-012

Type: Grammar

Difficulty: Medium

Bloom Level: Apply

Topic: Prepositions

Marks: 1

**Question:** Choose the correct preposition:

\*"A tadpole grows \_\_\_\_\_ a froglet."\*

- (A) into
- (B) off
- (C) from
- (D) onto

### Question 13

Question ID: CH11-GRA-013

Type: Grammar

Difficulty: Medium

Bloom Level: Apply

Topic: Conjunctions

Marks: 1

**Question:** Join the two sentences using a suitable conjunction:

\*"A tadpole lives in water. A tadpole breathes with gills."\*

- (A) A tadpole lives in water **\*\*and\*\*** breathes with gills.
- (B) A tadpole lives in water **\*\*but\*\*** breathes with gills.
- (C) A tadpole lives in water **\*\*unless\*\*** breathes with gills.
- (D) A tadpole lives in water **\*\*so\*\*** breathes with gills.

### Question 14

Question ID: CH11-GRA-014

Type: Grammar

Difficulty: Medium

Bloom Level: Apply

Topic: Conjunctions

Marks: 1

**Question:** Complete the sentence with an appropriate conjunction:

\*"A tadpole has a tail, \_\_\_\_\_ an adult frog has no tail."\*

- (A) but / whereas
- (B) because
- (C) unless
- (D) so

### Question 15

Question ID: CH11-GRA-015

Type: Grammar

Difficulty: Medium

Bloom Level: Understand

Topic: Adverbs

Marks: 1

**Question:** Identify the Adverb of Time in the sentence:

\*"Later, the tadpole develops legs."\*

- (A) tadpole
- (B) Later
- (C) develops
- (D) legs

### Question 16

Question ID: CH11-GRA-016

Type: Grammar

Difficulty: Medium

Bloom Level: Apply

Topic: Articles

Marks: 1

**Question:** Fill in the blank with the correct article:

\*"A froglet grows into \_\_\_\_\_ adult frog."\*

- (A) a
- (B) an
- (C) the
- (D) no article needed

### Question 17

Question ID: CH11-GRA-017

Type: Grammar

Difficulty: Medium

Bloom Level: Apply

Topic: Articles

Marks: 1

**Question:** Fill in the blank with the correct article:

\*"Female frogs lay eggs in \_\_\_\_\_ water of the pond."\*

- (A) a
- (B) an
- (C) the
- (D) no article needed

### Question 18

Question ID: CH11-GRA-018

Type: Grammar

Difficulty: Medium

Bloom Level: Understand

Topic: Subject-Verb Agreement

Marks: 1

**Question:** Choose the verb that agrees with the subject:

\*"Female frogs \_\_\_\_\_ eggs in water."\*

- (A) lay
- (B) lays
- (C) is laying
- (D) was layed

### Question 19

Question ID: CH11-GRA-019

Type: Grammar

Difficulty: Hard

Bloom Level: Analyze

Topic: Sentence Types

Marks: 1

**Question:** What type of sentence is: "A frog goes through four stages of life."

- (A) Declarative / Assertive
- (B) Interrogative
- (C) Imperative
- (D) Exclamatory

### Question 20

Question ID: CH11-GRA-020

Type: Grammar

Difficulty: Hard

Bloom Level: Analyze

Topic: Sentence Types

Marks: 1

**Question:** What type of sentence is: "How do tadpoles breathe in water?"

- (A) Declarative
- (B) Interrogative
- (C) Exclamatory
- (D) Imperative

### Question 21

Question ID: CH11-GRA-021

Type: Grammar

Difficulty: Medium

Bloom Level: Apply

Topic: Plurals

Marks: 1

**Question:** What is the plural form of the noun 'life' in 'lifecycles'?

- (A) Lifes
- (B) Lives
- (C) Lifess
- (D) Lifen

### Question 22

Question ID: CH11-GRA-022

Type: Grammar

Difficulty: Medium

Bloom Level: Apply

Topic: Plurals

Marks: 1

**Question:** What is the plural form of 'lung'?

- (A) Lungs
- (B) Lunges
- (C) Lungies
- (D) Lungae

### Question 23

Question ID: CH11-GRA-023

Type: Grammar

Difficulty: Hard

Bloom Level: Analyze

Topic: Negative Sentences

Marks: 1

**Question:** Change into a negative sentence without changing the meaning:

"A tadpole lives in water."

- (A) A tadpole does not live on land.
- (B) A tadpole does not live in water.
- (C) A tadpole does not live.
- (D) A tadpole lives on land.

### Question 24

Question ID: CH11-GRA-024

Type: Grammar

Difficulty: Hard

Bloom Level: Apply

Topic: Punctuation Check

Marks: 1

**Question:** Select the correctly punctuated sentence.

- (A) the four stages of a frogs life are egg tadpole froglet and adult frog
- (B) The four stages of a frog's life are: egg, tadpole, froglet, and adult frog.
- (C) "The four stages of a frog's life are egg tadpole froglet and adult frog."
- (D) The four stages of a frogs life are egg, tadpole, froglet and adult frog.

### Question 25

Question ID: CH11-GRA-025

Type: Grammar

Difficulty: Hard

Bloom Level: Apply

Topic: Possessive Nouns

Marks: 1

**Question:** Choose the correct possessive form:



\*"The lifecycle of a frog" = \_\_\_\_\_ \*

- (A) A frogs lifecycle
- (B) A frog's lifecycle
- (C) A frogs' lifecycle
- (D) A frog lifecycle

## Answer Key

- | \*\*CH11-GRA-001\*\* | \*\*(A)\*\* | 'Frog', 'stages', 'life' are Common Nouns. |
- | \*\*CH11-GRA-002\*\* | \*\*(B)\*\* | 'Gills' is a Common Noun referring to breathing organ. |
- | \*\*CH11-GRA-003\*\* | \*\*(A)\*\* | Active Voice: Female frogs (Subject) + lay (Verb) + eggs (Object). |
- | \*\*CH11-GRA-004\*\* | \*\*(A)\*\* | Passive Voice: Lungs (Object) + are used by + froglet. |
- | \*\*CH11-GRA-005\*\* | \*\*(B)\*\* | 'Four' states number of stages (Adjective of Number). |
- | \*\*CH11-GRA-006\*\* | \*\*(A)\*\* | Young -> Younger -> Youngest. |
- | \*\*CH11-GRA-007\*\* | \*\*(A)\*\* | Long -> Longer -> Longest. |
- | \*\*CH11-GRA-008\*\* | \*\*(B)\*\* | Plural subject (tadpoles) takes base verb: hatch. |
- | \*\*CH11-GRA-009\*\* | \*\*(B)\*\* | Past tense of grow is grew. |
- | \*\*CH11-GRA-010\*\* | \*\*(B)\*\* | 'It' is a Personal Pronoun referring to frog. |
- | \*\*CH11-GRA-011\*\* | \*\*(B)\*\* | Preposition 'in' water. |
- | \*\*CH11-GRA-012\*\* | \*\*(A)\*\* | Preposition 'into' for stage transformation. |
- | \*\*CH11-GRA-013\*\* | \*\*(A)\*\* | Conjunction 'and' joins two facts about tadpoles. |
- | \*\*CH11-GRA-014\*\* | \*\*(A)\*\* | Conjunction 'but' shows contrast between tail presence/absence. |
- | \*\*CH11-GRA-015\*\* | \*\*(B)\*\* | 'Later' indicates time of leg development (Adverb of time). |
- | \*\*CH11-GRA-016\*\* | \*\*(B)\*\* | Indefinite article 'an' before vowel sound 'adult'. |
- | \*\*CH11-GRA-017\*\* | \*\*(C)\*\* | Definite article 'the' before specific noun 'the water of the pond'. |
- | \*\*CH11-GRA-018\*\* | \*\*(A)\*\* | Plural subject 'Female frogs' takes plural verb 'lay'. |
- | \*\*CH11-GRA-019\*\* | \*\*(A)\*\* | Statement of fact (Declarative). |
- | \*\*CH11-GRA-020\*\* | \*\*(B)\*\* | Question ending with question mark (Interrogative). |
- | \*\*CH11-GRA-021\*\* | \*\*(B)\*\* | Life -> Lives. |
- | \*\*CH11-GRA-022\*\* | \*\*(A)\*\* | Lung -> Lungs. |
- | \*\*CH11-GRA-023\*\* | \*\*(A)\*\* | 'Does not live on land' preserves the meaning of lives in water. |
- | \*\*CH11-GRA-024\*\* | \*\*(B)\*\* | Apostrophe in frog's, colon, and commas are correct. |
- | \*\*CH11-GRA-025\*\* | \*\*(B)\*\* | Singular possessive: A frog's lifecycle. |

## 8. Extract & Passage Based Questions

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### ## Extract 1

> \*\*A frog goes through four stages of life. As the frog grows, it changes. This is called metamorphosis. The four stages of a frog's life are: Egg - Female frogs lay eggs. This is also called spawning. Tadpoles hatch from these eggs. It takes from one to three weeks for the eggs to hatch.\*\*

#### Question 1

Question ID: CH11-EXT-001

Type: Extract Based

Difficulty: Easy

Bloom Level: Remember

Topic: Number of Life Stages

Marks: 1

**Question:** How many stages of life does a frog go through according to Extract 1?

- (A) Two stages
- (B) Three stages
- (C) Four stages
- (D) Five stages

#### Question 2

Question ID: CH11-EXT-002

Type: Extract Based

Difficulty: Easy

Bloom Level: Remember

Topic: Metamorphosis Definition

Marks: 1

**Question:** What is the process called when a frog changes form as it grows?

- (A) Hibernation
- (B) Metamorphosis
- (C) Spawning
- (D) Hatching

#### Question 3

Question ID: CH11-EXT-003

Type: Extract Based

Difficulty: Easy

Bloom Level: Remember

Topic: Egg Laying Term

Marks: 1

**Question:** What is egg laying by female frogs called in Extract 1?

- (A) Metamorphosis
- (B) Spawning

- (C) Hopping
- (D) Breathing

#### Question 4

Question ID: CH11-EXT-004

Type: Extract Based

Difficulty: Easy

Bloom Level: Remember

Topic: Hatching Outcome & Duration

Marks: 1

**Question:** What hatches out of the frog eggs, and how long does hatching take?

- (A) Adult frogs in 1 day
- (B) Tadpoles in one to three weeks
- (C) Fish in 4 weeks
- (D) Froglets in 6 months

#### Question 5

Question ID: CH11-EXT-005

Type: Extract Based

Difficulty: Medium

Bloom Level: Understand

Topic: First Stage Description

Marks: 1

**Question:** Which stage of a frog's life cycle is described in detail in Extract 1?

- (A) The Egg stage
- (B) The Froglet stage
- (C) The Adult Frog stage
- (D) The Lizard stage

## Extract 2

> **"Tadpole** - A tadpole is born with a tail but no legs. The tadpole lives in water. It breathes with gills. Later on the tadpole develops legs. **Froglet** - A tadpole grows into a froglet. This froglet has developed legs for hopping. It now has lungs which help it breathe."

#### Question 6

Question ID: CH11-EXT-006

Type: Extract Based

Difficulty: Easy

Bloom Level: Remember

Topic: Tadpole Body Parts

Marks: 1

**Question:** What body features does a tadpole have when it is born according to Extract 2?

- (A) Legs but no tail
- (B) A tail but no legs

- (C) Wings and lungs
- (D) No tail and no legs

### Question 7

Question ID: CH11-EXT-007

Type: Extract Based

Difficulty: Easy

Bloom Level: Remember

Topic: Tadpole Habitat & Organ

Marks: 1

**Question:** Where does a tadpole live, and how does it breathe?

- (A) Lives on land; breathes with lungs
- (B) Lives in water; breathes with gills
- (C) Lives in trees; breathes with skin
- (D) Lives underground; breathes with nose

### Question 8

Question ID: CH11-EXT-008

Type: Extract Based

Difficulty: Easy

Bloom Level: Remember

Topic: Intermediate Stage Transformation

Marks: 1

**Question:** What does a tadpole grow into as legs and lungs develop?

- (A) Adult Frog
- (B) Froglet
- (C) Fish
- (D) Caterpillar

### Question 9

Question ID: CH11-EXT-009

Type: Extract Based

Difficulty: Medium

Bloom Level: Understand

Topic: Froglet Organs & Movement

Marks: 1

**Question:** What new organ and limbs does a froglet possess for breathing air and moving on land?

- (A) Gills and fins
- (B) Lungs and legs for hopping
- (C) Tail and wings
- (D) Shell and claws

### Question 10

Question ID: CH11-EXT-010

Type: Extract Based

Difficulty: Medium

Bloom Level: Understand

Topic: Respiratory Shift

Marks: 1

**Question:** How does breathing change when a tadpole grows into a froglet in Extract 2?

- (A) From breathing with gills to breathing with lungs
- (B) From breathing with lungs to breathing with gills
- (C) From breathing air to breathing water
- (D) Breathing stops completely

## Extract 3

> **"Frog - A froglet grows into an adult frog. An adult frog has no tail. A new life cycle begins when a female frog lays eggs."**

### Question 11

Question ID: CH11-EXT-011

Type: Extract Based

Difficulty: Easy

Bloom Level: Remember

Topic: Last Life Stage

Marks: 1

**Question:** What does a froglet grow into in Extract 3?

- (A) Tadpole
- (B) Adult frog
- (C) Fish
- (D) Bird

### Question 12

Question ID: CH11-EXT-012

Type: Extract Based

Difficulty: Easy

Bloom Level: Remember

Topic: Adult Frog Tail Feature

Marks: 1

**Question:** What is a key physical feature of an adult frog mentioned in Extract 3?

- (A) It has a long tail.
- (B) It has no tail.
- (C) It has no legs.
- (D) It has gills.

### Question 13

Question ID: CH11-EXT-013

Type: Extract Based

Difficulty: Easy

Bloom Level: Remember

Topic: New Lifecycle Start

Marks: 1

**Question:** When does a new life cycle of a frog begin?

- (A) When an adult frog loses its legs
- (B) When a female frog lays eggs
- (C) When a tadpole stops swimming
- (D) When winter starts

### Question 14

Question ID: CH11-EXT-014

Type: Extract Based

Difficulty: Medium

Bloom Level: Understand

Topic: Tail Comparison

Marks: 1

**Question:** How does the tail feature differ between a tadpole and an adult frog?

- (A) Tadpole has no tail; adult frog has a tail.
- (B) Tadpole has a tail; adult frog has no tail.
- (C) Both have long tails.
- (D) Neither has a tail.

### Question 15

Question ID: CH11-EXT-015

Type: Extract Based

Difficulty: Medium

Bloom Level: Understand

Topic: Continuous Cycle Concept

Marks: 1

**Question:** Why is this biological process described as a "cycle"?

- (A) Because it repeats continuously as adult female frogs lay new eggs.
- (B) Because frogs move in circles.
- (C) Because it lasts for 100 years.
- (D) Because it happens only once in nature.

## Short Comprehension Questions on Extracts

### Question 16

Question ID: CH11-EXT-016

Type: Extract Based (Short Answer)

*Difficulty: Medium*

*Bloom Level: Understand*

*Topic: Extract 1 Analysis*

*Marks: 2*

**Question:** Refer to Extract 1: Define metamorphosis and state how long it takes for frog eggs to hatch into tadpoles.

### **Question 17**

*Question ID: CH11-EXT-017*

*Type: Extract Based (Short Answer)*

*Difficulty: Medium*

*Bloom Level: Understand*

*Topic: Extract 2 Analysis*

*Marks: 2*

**Question:** Refer to Extract 2: Compare the habitat, limbs, and breathing organs of a tadpole with those of a froglet.

### **Question 18**

*Question ID: CH11-EXT-018*

*Type: Extract Based (Short Answer)*

*Difficulty: Medium*

*Bloom Level: Understand*

*Topic: Extract 3 Analysis*

*Marks: 2*

**Question:** Refer to Extract 3: Describe the final adult frog stage and explain how the life cycle restarts.

### **Question 19**

*Question ID: CH11-EXT-019*

*Type: Extract Based (Short Answer)*

*Difficulty: Hard*

*Bloom Level: Evaluate*

*Topic: Extract 1 & 2 Synthesis*

*Marks: 2*

**Question:** Explain why developing lungs and legs is essential for the transition from tadpole to froglet.

### **Question 20**

*Question ID: CH11-EXT-020*

*Type: Extract Based (Short Answer)*

*Difficulty: Hard*

*Bloom Level: Evaluate*

*Topic: Synthesis Across Extracts*

*Marks: 2*

**Question:** Trace the complete physical transformation of a frog across all four stages as described in Extracts 1, 2, and 3.

**## Additional Comprehension Questions (21 to 25)**

### Question 21

Question ID: CH11-EXT-021

Type: Extract Based

Difficulty: Medium

Bloom Level: Apply

Topic: Vocabulary Matching

Marks: 1

**Question:** Which word in Extract 1 means **'coming out of an egg'**?

- (A) Spawning
- (B) Hatch
- (C) Metamorphosis
- (D) Stage

### Question 22

Question ID: CH11-EXT-022

Type: Extract Based

Difficulty: Easy

Bloom Level: Remember

Topic: Gender Identification

Marks: 1

**Question:** Which gender of frogs lays eggs according to Extract 1?

- (A) Male frogs
- (B) Female frogs
- (C) Young tadpoles
- (D) Both male and female

### Question 23

Question ID: CH11-EXT-023

Type: Extract Based

Difficulty: Hard

Bloom Level: Analyze

Topic: Hopping Adaptation Purpose

Marks: 1

**Question:** Why does Extract 2 state that a froglet develops legs for hopping?

- (A) To swim faster in water.
- (B) To enable land movement as it transitions out of the pond environment.
- (C) To lay eggs.
- (D) To breathe underwater.

### Question 24

Question ID: CH11-EXT-024

Type: Extract Based

Difficulty: Medium



*Bloom Level: Apply*

*Topic: Antonym Identification*

*Marks: 1*

**Question:** What is the antonym of **'adult'** in Extract 3?

- (A) Mature
- (B) Young / Juvenile
- (C) Female
- (D) Large

## Question 25

*Question ID: CH11-EXT-025*

*Type: Extract Based (Short Answer)*

*Difficulty: Hard*

*Bloom Level: Evaluate*

*Topic: Chapter Synthesis*

*Marks: 2*

**Question:** Summarize the biological journey of a frog from a tiny egg floating in water to a hopping adult frog without a tail.

## Answer Key

| **CH11-EXT-001** | **(C)** | Four stages of life. |

| **CH11-EXT-002** | **(B)** | Metamorphosis. |

| **CH11-EXT-003** | **(B)** | Spawning. |

| **CH11-EXT-004** | **(B)** | Tadpoles hatch in 1 to 3 weeks. |

| **CH11-EXT-005** | **(A)** | The Egg stage (spawning and hatching). |

| **CH11-EXT-006** | **(B)** | A tail but no legs. |

| **CH11-EXT-007** | **(B)** | Lives in water; breathes with gills. |

| **CH11-EXT-008** | **(B)** | Grows into a froglet. |

| **CH11-EXT-009** | **(B)** | Lungs and legs for hopping. |

| **CH11-EXT-010** | **(A)** | From gills (water) to lungs (air). |

| **CH11-EXT-011** | **(B)** | Adult frog. |

| **CH11-EXT-012** | **(B)** | An adult frog has no tail. |

| **CH11-EXT-013** | **(B)** | When a female frog lays eggs. |

| **CH11-EXT-014** | **(B)** | Tadpole has a tail; adult frog has no tail. |

| **CH11-EXT-015** | **(A)** | Repeats continuously as female frogs lay new eggs. |

| **CH11-EXT-016** | **Model Answer**: Metamorphosis is the process of changing form as a frog grows. It takes 1 to 3 weeks for frog eggs to hatch into tadpoles. |

| **CH11-EXT-017** | **Model Answer**: A tadpole lives in water, has a tail with no legs, and breathes through gills. A froglet has developed legs for hopping, lungs for breathing air, and a shrinking tail. |

| **CH11-EXT-018** | **Model Answer**: In the final stage, the froglet becomes an adult frog with no tail. The life cycle restarts when a female adult frog lays new eggs in water. |

| **CH11-EXT-019** | **Model Answer**: Gills only extract oxygen from water. Developing lungs and legs allows the froglet to leave the pond and move onto land to find food and survive. |

| **CH11-EXT-020** | **Model Answer**: It starts as an Egg  $\rightarrow$  hatches into a water-dwelling Tadpole with a tail and gills  $\rightarrow$  develops legs and lungs as a Froglet  $\rightarrow$  loses its tail completely to become an Adult Frog. |

| **CH11-EXT-021** | **(B)** | Hatch means coming out of an egg. |

| **CH11-EXT-022** | **(B)** | Female frogs lay eggs. |

| **CH11-EXT-023** | **(B)** | To enable land movement during transition out of pond. |

| **CH11-EXT-024** | **(B)** | Antonym of adult is young or juvenile. |

| **CH11-EXT-025** | **Model Answer**: A female frog lays eggs (\*spawning\*) which hatch in 1-3 weeks into aquatic tadpoles with gills and tails. As legs and lungs develop, it becomes a hopping froglet and finally a tailless adult frog capable of living on land. |

## 9. Case & Scenario Based Questions

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### ## Scenario 1: Pond Exploration Observation

> Rahul visited a school nature pond in March. He saw a jelly-like mass floating near water plants. Three weeks later, he noticed hundreds of tiny black creatures swimming with long tails but no legs.

#### Question 1

Question ID: CH11-CAS-001

Type: Case Based

Difficulty: Easy

Bloom Level: Apply

Topic: Stage Identification

Marks: 1

**Question:** Which two life stages of a frog did Rahul observe in the pond?

- (A) Adult Frog and Froglet
- (B) Egg (jelly mass) and Tadpole (swimming with tails)
- (C) Fish and Caterpillar
- (D) Lizard and Turtle

#### Question 2

Question ID: CH11-CAS-002

Type: Case Based

Difficulty: Medium

Bloom Level: Evaluate

Topic: Environmental Protection

Marks: 1

**Question:** Why is it important for Rahul and his classmates not to disturb the floating jelly egg mass in the pond?

- (A) Because eggs dry out easily and need clean, undisturbed water to hatch safely into tadpoles.
- (B) Because eggs are poisonous.
- (C) Because eggs are expensive.
- (D) To keep the water dark.

### ## Scenario 2: Pollution in Freshwater Ponds

> A factory dumped chemical soap waste into a local pond where female frogs had laid eggs. Shortly after, science students noticed that very few tadpoles hatched, and those that hatched struggled to breathe.

#### Question 3

Question ID: CH11-CAS-003

Type: Case Based

Difficulty: Medium

Bloom Level: Apply

Topic: Gills Sensitivity

Marks: 1

**Question:** Why are tadpoles especially vulnerable to chemical water pollution compared to adult frogs?

- (A) Because tadpoles eat chemicals.
- (B) Because tadpoles live entirely in water and breathe through delicate gills, which absorb toxic pollutants directly.
- (C) Because tadpoles have no eyes.
- (D) Because tadpoles can fly away.

#### Question 4

Question ID: CH11-CAS-004

Type: Case Based

Difficulty: Hard

Bloom Level: Evaluate

Topic: Ecological Conservation

Marks: 1

**Question:** How can a local community protect frog lifecycles in neighborhood ponds?

- (A) By draining all pond water.
- (B) By preventing chemical dumping, keeping pond water clean, and protecting aquatic vegetation.
- (C) By introducing plastic fish.
- (D) By building concrete walls around ponds.

## Scenario 3: Adaptation from Water to Land

> Maya watched a creature in her garden pond. Last month it swam continuously using a tail. Today, it has four legs, a tiny shrinking tail stump, and hopped out of the water onto a floating lily pad to catch a fly.

#### Question 5

Question ID: CH11-CAS-005

Type: Case Based

Difficulty: Medium

Bloom Level: Apply

Topic: Froglet Stage Features

Marks: 1

**Question:** Which stage of metamorphosis is Maya observing in her garden?

- (A) Egg stage
- (B) Froglet stage (developing legs, lungs, and shrinking tail)
- (C) Tadpole stage
- (D) Adult toad stage

## Scenario 4: Natural Insect Control

> A farmer noticed a high population of crop-destroying insects on his farm. Instead of spraying toxic chemical pesticides, he built a small freshwater pond nearby, which attracted adult frogs. Within a month, the insect population dropped significantly.

#### Question 6

Question ID: CH11-CAS-006

Type: Case Based

Difficulty: Easy

Bloom Level: Understand

Topic: Ecological Value of Adult Frogs

Marks: 1

**Question:** How did attracting adult frogs help the farmer control pests naturally?

- (A) Adult frogs eat plant roots.
- (B) Adult frogs are carnivorous land animals that catch and eat large quantities of agricultural insects.
- (C) Adult frogs make loud noise.
- (D) Adult frogs scare birds.

## Multiple-Choice Case Scenarios (7 to 20)

### Question 7

Question ID: CH11-CAS-007

Type: Case Based

Difficulty: Easy

Bloom Level: Apply

Topic: Hatching Time Calculation

Marks: 1

**Question:** A female frog laid eggs on April 1st. According to Chapter 11, between which dates should tadpoles start hatching?

- (A) April 2nd to April 3rd (1-2 days)
- (B) April 8th to April 22nd (1 to 3 weeks)
- (C) June 1st to July 1st (2-3 months)
- (D) Next year

### Question 8

Question ID: CH11-CAS-008

Type: Case Based

Difficulty: Medium

Bloom Level: Understand

Topic: Organ Shift Identification

Marks: 1

**Question:** A student compares a tadpole's breathing organ with a froglet's breathing organ. What shift occurred?

- (A) From lungs to gills
- (B) From gills (underwater breathing) to lungs (air breathing)
- (C) From skin to wings
- (D) No change

### Question 9

Question ID: CH11-CAS-009

Type: Case Based

*Difficulty: Hard*

*Bloom Level: Analyze*

*Topic: Tailless Feature of Adult*

*Marks: 1*

**Question:** An adult frog is jumping across lawn grass. A student notices it has no tail. What happened to the tail it had as a tadpole?

- (A) It fell off in the mud.
- (B) The tail was gradually resorbed by the body as a source of energy during the froglet stage.
- (C) It was bitten by a fish.
- (D) It turned into a leg.

### **Question 10**

*Question ID: CH11-CAS-010*

*Type: Case Based*

*Difficulty: Medium*

*Bloom Level: Apply*

*Topic: Spawning Location Choice*

*Marks: 1*

**Question:** Why do female frogs always seek water bodies (ponds, streams) for laying eggs rather than dry soil?

- (A) Because frog eggs have soft jelly coatings that require moisture to survive and hatch.
- (B) Because frogs cannot walk on soil.
- (C) To hide from sunshine.
- (D) Because eggs float in air.

### **Question 11**

*Question ID: CH11-CAS-011*

*Type: Case Based*

*Difficulty: Hard*

*Bloom Level: Evaluate*

*Topic: Amphibian Definition Application*

*Marks: 1*

**Question:** A biology quiz asks: *"Why is a frog classified as an amphibian while a fish is not?"*

- (A) Because fish have scales.
- (B) Because a frog undergoes metamorphosis from a water-dwelling gill-breather to a land-dwelling lung-breather, living in both environments.
- (C) Because fish live on land.
- (D) Because frogs lay eggs.

### **Question 12**

*Question ID: CH11-CAS-012*

*Type: Case Based*

*Difficulty: Medium*

*Bloom Level: Apply*

*Topic: Classroom Observation Ethics*

*Marks: 1*

**Question:** When observing tadpoles in a school aquarium, what is the proper ethical practice once froglets develop lungs and legs?

- (A) Keep them locked in water forever.
- (B) Provide a floating land platform or gently release them back into their natural pond habitat.
- (C) Throw them in dry soil.
- (D) Stop feeding them.

### Question 13

*Question ID: CH11-CAS-013*

*Type: Case Based*

*Difficulty: Hard*

*Bloom Level: Evaluate*

*Topic: Diet Transition Analysis*

*Marks: 1*

**Question:** Tadpoles feed mainly on pond algae, whereas adult frogs catch flying insects. What advantage does this dietary difference offer?

- (A) It causes food shortages.
- (B) It prevents competition for food between young tadpoles and adult frogs living in the same area.
- (C) Tadpoles dislike insects.
- (D) Adult frogs cannot swim.

### Question 14

*Question ID: CH11-CAS-014*

*Type: Case Based*

*Difficulty: Medium*

*Bloom Level: Understand*

*Topic: Definition of Spawning*

*Marks: 1*

**Question:** A wildlife documentary shows thousands of frogs congregating in shallow water during monsoon rains to lay eggs. What term describes this mass egg laying?

- (A) Hibernation
- (B) Spawning
- (C) Migration
- (D) Molting

### Question 15

*Question ID: CH11-CAS-015*

*Type: Case Based*

*Difficulty: Hard*

*Bloom Level: Evaluate*

*Topic: Climate Change Impact on Ponds*

*Marks: 1*

**Question:** If a severe summer drought causes a pond to dry up completely within 5 days of egg laying, what will happen to the frog lifecycle?

- (A) Eggs will hatch into land frogs immediately.
- (B) The eggs will dry out and die before the 1-3 week hatching period, disrupting the local population.
- (C) Tadpoles will learn to fly.
- (D) Adult frogs will turn into fish.

### **Question 16**

*Question ID: CH11-CAS-016*

*Type: Case Based*

*Difficulty: Medium*

*Bloom Level: Apply*

*Topic: School Yard Bio-Pond Design*

*Marks: 1*

**Question:** A school eco-club wants to create a frog-friendly bio-pond. What two essential zones must the pond feature?

- (A) Deep water zone for tadpoles and a shallow muddy/plant land zone for froglets and adult frogs.
- (B) Salt water and ice.
- (C) Concrete walls with no plants.
- (D) High-speed waterfalls only.

### **Question 17**

*Question ID: CH11-CAS-017*

*Type: Case Based*

*Difficulty: Easy*

*Bloom Level: Remember*

*Topic: Leg Function for Froglet*

*Marks: 1*

**Question:** What specific movement are the newly developed legs of a froglet used for on land?

- (A) Flying
- (B) Hopping
- (C) Running on two feet
- (D) Crawling like a snake

### **Question 18**

*Question ID: CH11-CAS-018*

*Type: Case Based*

*Difficulty: Medium*

*Bloom Level: Apply*

*Topic: Respiratory Adaptation in Froglet*

*Marks: 1*



**Question:** A froglet spends time near the water's edge, occasionally sticking its head above water. What organ is it using to breathe atmospheric air?

- (A) Gills
- (B) Lungs
- (C) Tail
- (D) Fins

### Question 19

*Question ID: CH11-CAS-019*

*Type: Case Based*

*Difficulty: Hard*

*Bloom Level: Analyze*

*Topic: Metamorphosis Meaning Recall*

*Marks: 1*

**Question:** A student defines metamorphosis as *"changing form from one stage to another"*. Which animal from Chapter 11 demonstrates this process best?

- (A) Dog
- (B) Frog
- (C) Horse
- (D) Elephant

### Question 20

*Question ID: CH11-CAS-020*

*Type: Case Based*

*Difficulty: Easy*

*Bloom Level: Remember*

*Topic: Slogan Identification*

*Marks: 1*

**Question:** Which slogan best captures the essence of Chapter 11?

- (A) "From egg to tadpole, froglet to frog—protect life cycles in every pond and bog!"
- (B) "Frogs should live in dry deserts!"
- (C) "Tadpoles do not need water!"
- (D) "Metamorphosis means losing all legs!"

## Short Scenario Analysis Questions (21 to 25)

### Question 21

*Question ID: CH11-CAS-021*

*Type: Case Based (Short Answer)*

*Difficulty: Medium*

*Bloom Level: Apply*

*Topic: Pond Care Actions*

*Marks: 2*

**Question:** Mention two practical actions school students can take to protect frog eggs and tadpoles in local ponds.

### Question 22

Question ID: CH11-CAS-022

Type: Case Based (Short Answer)

Difficulty: Hard

Bloom Level: Analyze

Topic: Organ Transition Analysis

Marks: 2

**Question:** Analyze how the transition from gills to lungs enables a frog to explore new habitats on land.

### Question 23

Question ID: CH11-CAS-023

Type: Case Based (Short Answer)

Difficulty: Medium

Bloom Level: Apply

Topic: Environmental Indicators

Marks: 2

**Question:** Why do scientists consider frogs "environmental indicators" for water cleanliness in an ecosystem?

### Question 24

Question ID: CH11-CAS-024

Type: Case Based (Short Answer)

Difficulty: Hard

Bloom Level: Evaluate

Topic: Tail Absorption Benefit

Marks: 2

**Question:** Explain why absorbing its own tail provides a double benefit (nutrition and streamlined land body) to a growing froglet.

### Question 25

Question ID: CH11-CAS-025

Type: Case Based (Short Answer)

Difficulty: Hard

Bloom Level: Create

Topic: Educational Tribute Slogan

Marks: 2

**Question:** Create a 2-line rhyming tribute slogan honoring the 4-stage metamorphosis of a frog from water to land.

## Answer Key & Solutions

| **\*\*CH11-CAS-001\*\*** | **\*\* (B) \*\*** | Egg (jelly mass) and Tadpole (swimming with tails). |

| **\*\*CH11-CAS-002\*\*** | **\*\* (A) \*\*** | Eggs dry out easily and need clean, undisturbed water to hatch safely. |

| **\*\*CH11-CAS-003\*\*** | **\*\* (B) \*\*** | Tadpoles live in water and breathe through delicate gills absorbing pollutants. |

| **\*\*CH11-CAS-004\*\*** | **\*\* (B) \*\*** | Prevent chemical dumping, keep water clean, protect aquatic vegetation. |

- | **\*\*CH11-CAS-005\*\*** | **\*\* (B) \*\*** | Froglet stage (developing legs, lungs, and shrinking tail). |
- | **\*\*CH11-CAS-006\*\*** | **\*\* (B) \*\*** | Adult frogs are carnivorous land animals that catch agricultural insects. |
- | **\*\*CH11-CAS-007\*\*** | **\*\* (B) \*\*** | Takes 1 to 3 weeks (April 8th to April 22nd). |
- | **\*\*CH11-CAS-008\*\*** | **\*\* (B) \*\*** | From gills (underwater) to lungs (air breathing). |
- | **\*\*CH11-CAS-009\*\*** | **\*\* (B) \*\*** | Tail was resorbed by the body as a source of energy during froglet stage. |
- | **\*\*CH11-CAS-010\*\*** | **\*\* (A) \*\*** | Frog eggs have soft jelly coatings requiring water moisture to survive. |
- | **\*\*CH11-CAS-011\*\*** | **\*\* (B) \*\*** | Undergoes metamorphosis from water gill-breather to land lung-breather. |
- | **\*\*CH11-CAS-012\*\*** | **\*\* (B) \*\*** | Provide floating land platform or release them into natural pond habitat. |
- | **\*\*CH11-CAS-013\*\*** | **\*\* (B) \*\*** | Prevents food competition between young tadpoles and adult frogs. |
- | **\*\*CH11-CAS-014\*\*** | **\*\* (B) \*\*** | Spawning (laying egg mass in water). |
- | **\*\*CH11-CAS-015\*\*** | **\*\* (B) \*\*** | Eggs dry out and die before the 1-3 week hatching period. |
- | **\*\*CH11-CAS-016\*\*** | **\*\* (A) \*\*** | Deep water zone for tadpoles and shallow land zone for froglets/adults. |
- | **\*\*CH11-CAS-017\*\*** | **\*\* (B) \*\*** | Hopping on land. |
- | **\*\*CH11-CAS-018\*\*** | **\*\* (B) \*\*** | Lungs (developed for breathing atmospheric oxygen). |
- | **\*\*CH11-CAS-019\*\*** | **\*\* (B) \*\*** | Frog demonstrates complete metamorphosis across 4 stages. |
- | **\*\*CH11-CAS-020\*\*** | **\*\* (A) \*\*** | "From egg to tadpole, froglet to frog—protect life cycles in every pond and bog!" |
- | **\*\*CH11-CAS-021\*\*** | **\*\*Model Answer\*\***: 1. Avoid dumping trash, plastic, or chemicals into pond water.  
2. Protect pond plants where female frogs attach their egg masses. |
- | **\*\*CH11-CAS-022\*\*** | **\*\*Model Answer\*\***: Gills can only take oxygen dissolved in water. Lungs allow the froglet to breathe atmospheric air, enabling it to leave overcrowded ponds and live on land. |
- | **\*\*CH11-CAS-023\*\*** | **\*\*Model Answer\*\***: Because frog skin and tadpole gills easily absorb water pollutants, a healthy frog population indicates clean, unpolluted water in an ecosystem. |
- | **\*\*CH11-CAS-024\*\*** | **\*\*Model Answer\*\***: Absorbing the tail supplies rich energy while the froglet adapts its mouth, while removing an unnecessary swimming tail creates a lightweight body for land hopping. |
- | **\*\*CH11-CAS-025\*\*** | **\*\*Model Slogan\*\***:  
\*From swimming tadpole with a tail,  
To land-hopping frog, nature's wonder prevails!\*

## 10. Picture & Visual Based Questions

### ## Scene 1: Circular Lifecycle Diagram

> **Visual Description:** A circular scientific diagram illustrating the four sequential stages of a frog's life cycle. Arrows connect Stage 1: Jelly-like **Egg** mass in water  $\rightarrow$  Stage 2: Swimming **Tadpole** with tail and gills  $\rightarrow$  Stage 3: **Froglet** with developing hind legs, lungs, and short tail  $\rightarrow$  Stage 4: **Adult Frog** on land with no tail laying eggs to complete the circle.

#### Question 1

Question ID: CH11-PIC-001

Type: Picture Based

Difficulty: Easy

Bloom Level: Remember

Topic: Lifecycle Diagram Identification

Marks: 1

**Question:** What scientific process is illustrated in this circular 4-stage diagram?

- (A) Plant photosynthesis
- (B) Lifecycle / Metamorphosis of a Frog
- (C) Water cycle
- (D) Bird migration

#### Question 2

Question ID: CH11-PIC-002

Type: Picture Based

Difficulty: Easy

Bloom Level: Remember

Topic: Stage 1 Identification

Marks: 1

**Question:** What is the first stage labeled in the diagram where female frogs lay eggs in water?

- (A) Tadpole
- (B) Egg (Spawning)
- (C) Froglet
- (D) Adult Frog

#### Question 3

Question ID: CH11-PIC-003

Type: Picture Based

Difficulty: Hard

Bloom Level: Evaluate

Topic: Circular Arrow Meaning

Marks: 1

**Question:** Why are the four stages in Scene 1 arranged in a continuous circular arrow loop?

- (A) Because frogs run in circles.

- (B) It shows that the life cycle repeats perpetually as adult female frogs lay new eggs.
- (C) To make the diagram look like a wheel.
- (D) By accident.

### ## Scene 2: Swimming Tadpole Stage

> **Visual Description**: A detailed close-up illustration of Stage 2 (Tadpole) swimming in pond water. The tadpole has a large round head, a long flexible swimming tail, small external gill tufts on the side of its head, and no legs. Small green water algae float nearby.

### Question 4

Question ID: CH11-PIC-004

Type: Picture Based

Difficulty: Easy

Bloom Level: Remember

Topic: Tadpole Identification

Marks: 1

**Question:** Which stage of the frog's lifecycle is depicted swimming underwater in Scene 2?

- (A) Egg
- (B) Tadpole
- (C) Froglet
- (D) Adult Frog

### Question 5

Question ID: CH11-PIC-005

Type: Picture Based

Difficulty: Easy

Bloom Level: Remember

Topic: Anatomical Features

Marks: 1

**Question:** Which two anatomical features are clearly visible on the tadpole in Scene 2?

- (A) Four legs and lungs
- (B) A long tail and gills (with no legs)
- (C) Wings and beak
- (D) Shell and claws

### Question 6

Question ID: CH11-PIC-006

Type: Picture Based

Difficulty: Hard

Bloom Level: Analyze

Topic: Underwater Adaptation

Marks: 1

**Question:** How does the tail structure in Scene 2 help the tadpole survive underwater?

- (A) It holds food.

- (B) Its long, flexible fin-like shape allows fast sideways wiggling propulsion through water.
- (C) It acts as an anchor.
- (D) It helps it hop on grass.

### ## Scene 3: Transition Stage - The Froglet

> **Visual Description**: An illustration of Stage 3 (Froglet) resting on a floating pond leaf near the water edge. The froglet has developed four legs (strong hind legs for hopping), visible nostrils/lungs for air breathing, but still retains a small shrinking tail stump at the back.

### Question 7

Question ID: CH11-PIC-007

Type: Picture Based

Difficulty: Easy

Bloom Level: Remember

Topic: Froglet Identification

Marks: 1

**Question:** Which intermediate stage is depicted in Scene 3 featuring developed legs alongside a shrinking tail stump?

- (A) Egg
- (B) Tadpole
- (C) Froglet
- (D) Adult Frog

### Question 8

Question ID: CH11-PIC-008

Type: Picture Based

Difficulty: Medium

Bloom Level: Understand

Topic: Habitat Transition

Marks: 1

**Question:** Where is the froglet positioned in Scene 3, reflecting its dual adaptation?

- (A) Deep at the bottom of the ocean
- (B) At the water edge resting on a lily pad between water and land
- (C) High in a tree top
- (D) Buried under dry sand

### Question 9

Question ID: CH11-PIC-009

Type: Picture Based

Difficulty: Hard

Bloom Level: Evaluate

Topic: Tail Stump Resorption Visual

Marks: 1

**Question:** What is happening to the small tail stump visible behind the froglet in Scene 3?

- (A) It is growing longer into a snake tail.
- (B) It is being gradually resorbed by the body as a source of nourishment until it disappears in the adult frog.
- (C) It turned into a fifth leg.
- (D) It is falling off into the pond.

#### ## Scene 4: Tailless Adult Frog on Land

> **Visual Description**: An illustration of Stage 4 (Adult Frog) sitting on green grass by the pond bank. It has smooth moist skin, large bulging eyes, powerful folded hind legs ready for hopping, a wide mouth catching a flying dragon fly, and **no tail**.

### Question 10

Question ID: CH11-PIC-010

Type: Picture Based

Difficulty: Easy

Bloom Level: Remember

Topic: Adult Frog Identification

Marks: 1

**Question:** Which final stage of the lifecycle is shown sitting on land in Scene 4?

- (A) Tadpole
- (B) Egg mass
- (C) Froglet
- (D) Adult Frog

### Question 11

Question ID: CH11-PIC-011

Type: Picture Based

Difficulty: Easy

Bloom Level: Remember

Topic: Key Difference from Tadpole

Marks: 1

**Question:** What key physical feature distinguishes the adult frog in Scene 4 from the tadpole in Scene 2?

- (A) The adult frog has a tail.
- (B) The adult frog has four legs and no tail.
- (C) The adult frog has gills.
- (D) The adult frog lives underwater only.

### Question 12

Question ID: CH11-PIC-012

Type: Picture Based

Difficulty: Hard

Bloom Level: Evaluate

Topic: Hind Leg Folded Position

Marks: 1

**Question:** Why are the adult frog's long hind legs shown folded tightly like Z-springs in Scene 4?

- (A) Because it is sleeping.
- (B) The coiled position stores muscle energy for explosive, high-speed leaps/hops across land to catch prey or escape danger.
- (C) Because its legs are broken.
- (D) To keep warm.

## Additional Visual & Observation Questions (13 to 20)

### Question 13

Question ID: CH11-PIC-013

Type: Picture Based

Difficulty: Medium

Bloom Level: Apply

Topic: Image Captioning

Marks: 1

**Question:** Which caption best fits Scene 1 (circular 4-stage diagram)?

- (A) Fish Species of World
- (B) The 4-Stage Metamorphosis: Lifecycle of a Frog from Egg to Adult
- (C) Water Conservation in Ponds
- (D) How Birds Fly

### Question 14

Question ID: CH11-PIC-014

Type: Picture Based

Difficulty: Easy

Bloom Level: Remember

Topic: Jelly Mass Visual

Marks: 1

**Question:** An illustration shows a clump of translucent jelly spheres with black dots floating in water. What is this clump called?

- (A) Frog egg mass / Frogspawn
- (B) Fish scales
- (C) River pebbles
- (D) Water lilies

### Question 15

Question ID: CH11-PIC-015

Type: Picture Based

Difficulty: Hard

Bloom Level: Analyze

Topic: Chronological Sequence of Images

Marks: 1

**Question:** Arrange these 4 visual stages of a frog in correct chronological order:



\*(1) Adult frog on grass with no tail; (2) Female laying jelly eggs in water; (3) Froglet with legs and small tail; (4) Swimming tadpole with gills and long tail.\*

- (A) (2) -> (4) -> (3) -> (1)
- (B) (1) -> (2) -> (3) -> (4)
- (C) (4) -> (3) -> (2) -> (1)
- (D) (3) -> (4) -> (1) -> (2)

### Question 16

Question ID: CH11-PIC-016

Type: Picture Based

Difficulty: Medium

Bloom Level: Apply

Topic: Gill Tufts Visual

Marks: 1

**Question:** An illustration shows small delicate feathery tufts on the side of a tadpole's head. What organ do these feathery tufts represent?

- (A) Lungs
- (B) External Gills for underwater oxygen absorption
- (C) Ears
- (D) Fins

### Question 17

Question ID: CH11-PIC-017

Type: Picture Based

Difficulty: Hard

Bloom Level: Evaluate

Topic: Tongue Catching Insect Visual

Marks: 1

**Question:** An illustration shows an adult frog shooting out a long, sticky tongue to snag a fly. What does this feeding adaptation demonstrate?

- (A) Herbivorous plant eating
- (B) Carnivorous land-hunting adaptation enabled by lungs and strong land legs
- (C) Tadpole feeding
- (D) Filter feeding underwater

### Question 18

Question ID: CH11-PIC-018

Type: Picture Based

Difficulty: Medium

Bloom Level: Understand

Topic: Hatching Sequence Visual

Marks: 1

**Question:** An illustration shows a tiny black tadpole wriggling out of a soft jelly egg sphere. What word describes this action?

- (A) Spawning
- (B) Hatching
- (C) Hopping
- (D) Metamorphosis

### Question 19

Question ID: CH11-PIC-019

Type: Picture Based

Difficulty: Hard

Bloom Level: Analyze

Topic: Organ Shift Diagram Visual

Marks: 1

**Question:** A graphic diagram shows: **Gills (Water Breathing)  $\rightarrow$  Lungs (Air Breathing)**. Which transition stage in the frog's lifecycle does this organ shift mark?

- (A) Egg to Tadpole
- (B) Tadpole to Froglet
- (C) Adult Frog to Egg
- (D) Egg to Adult

### Question 20

Question ID: CH11-PIC-020

Type: Picture Based

Difficulty: Medium

Bloom Level: Apply

Topic: Question Framing Prompt

Marks: 1

**Question:** Look at Scene 4 (Adult Frog on land). Frame a question using the prompt **"What organ does an adult..."**.

- (A) What organ does an adult frog use to breathe air on land?
- (B) Where are the fish swimming?
- (C) How many weeks does it take to hatch?
- (D) Why are eggs black?

## Short Visual Description Tasks (21 to 25)

### Question 21

Question ID: CH11-PIC-021

Type: Picture Based (Short Answer)

Difficulty: Medium

Bloom Level: Create

Topic: Visual Scene Description

Marks: 2

**Question:** Write a 2-sentence description of an illustration showing the four stages of a frog's life cycle arranged in a circular diagram.

### Question 22

Question ID: CH11-PIC-022

Type: Picture Based (Short Answer)

Difficulty: Medium

Bloom Level: Apply

Topic: Science Poster Layout Design

Marks: 2

**Question:** If you were drawing a science poster titled "From Water to Land: A Frog's Metamorphosis", list four key visual elements you would draw.

### Question 23

Question ID: CH11-PIC-023

Type: Picture Based (Short Answer)

Difficulty: Hard

Bloom Level: Analyze

Topic: Visual Contrast - Tadpole vs Adult Frog

Marks: 2

**Question:** Explain why illustrating a swimming tadpole next to a hopping adult frog creates a striking visual demonstration of metamorphosis.

### Question 24

Question ID: CH11-PIC-024

Type: Picture Based (Short Answer)

Difficulty: Medium

Bloom Level: Create

Topic: Comic Strip Dialogue

Marks: 2

**Question:** Write speech bubbles for a 2-panel comic strip:

(Panel 1) Swimming Tadpole: **"I swim in the pond with my long tail and breathe with gills!"**

(Panel 2) Hopping Froglet: **"Look at my new legs and lungs! Soon my tail will be gone and I'll hop on land!"**

### Question 25

Question ID: CH11-PIC-025

Type: Picture Based (Short Answer)

Difficulty: Hard

Bloom Level: Evaluate

Topic: Cover Art Analysis

Marks: 2

**Question:** What key visual elements should be featured on the cover page of **"Chapter 11: Lifecycle of a Frog"** to make it scientifically accurate, colorful, and engaging for ICSE Class IV students?

## Answer Key & Solutions

| **\*\*CH11-PIC-001\*\*** | **\*\*(B)\*\*** | **Lifecycle / Metamorphosis of a Frog.** |

| **\*\*CH11-PIC-002\*\*** | **\*\* (B) \*\*** | Egg (Spawning). |

| **\*\*CH11-PIC-003\*\*** | **\*\* (B) \*\*** | Life cycle repeats perpetually as female frogs lay new eggs. |

| **\*\*CH11-PIC-004\*\*** | **\*\* (B) \*\*** | Tadpole. |

| **\*\*CH11-PIC-005\*\*** | **\*\* (B) \*\*** | A long tail and gills (no legs). |

| **\*\*CH11-PIC-006\*\*** | **\*\* (B) \*\*** | Long, flexible fin-like shape allows fast wiggling propulsion. |

| **\*\*CH11-PIC-007\*\*** | **\*\* (C) \*\*** | Froglet (legs developed + shrinking tail). |

| **\*\*CH11-PIC-008\*\*** | **\*\* (B) \*\*** | At water edge on a lily pad between water and land. |

| **\*\*CH11-PIC-009\*\*** | **\*\* (B) \*\*** | Gradually resorbed by the body as a source of energy. |

| **\*\*CH11-PIC-010\*\*** | **\*\* (D) \*\*** | Adult Frog. |

| **\*\*CH11-PIC-011\*\*** | **\*\* (B) \*\*** | Adult frog has four legs and no tail. |

| **\*\*CH11-PIC-012\*\*** | **\*\* (B) \*\*** | Coiled position stores energy for explosive hopping. |

| **\*\*CH11-PIC-013\*\*** | **\*\* (B) \*\*** | The 4-Stage Metamorphosis: Lifecycle of a Frog. |

| **\*\*CH11-PIC-014\*\*** | **\*\* (A) \*\*** | Frog egg mass / Frogspawn. |

| **\*\*CH11-PIC-015\*\*** | **\*\* (A) \*\*** | (2) Egg laying -> (4) Swimming tadpole -> (3) Froglet -> (1) Adult frog. |

| **\*\*CH11-PIC-016\*\*** | **\*\* (B) \*\*** | External Gills for underwater oxygen absorption. |

| **\*\*CH11-PIC-017\*\*** | **\*\* (B) \*\*** | Carnivorous land-hunting adaptation enabled by lungs and legs. |

| **\*\*CH11-PIC-018\*\*** | **\*\* (B) \*\*** | Hatching (wriggling out of egg sphere). |

| **\*\*CH11-PIC-019\*\*** | **\*\* (B) \*\*** | Tadpole to Froglet (lungs replace gills). |

| **\*\*CH11-PIC-020\*\*** | **\*\* (A) \*\*** | "What organ does an adult frog use to breathe air on land?" |

| **\*\*CH11-PIC-021\*\*** | **\*\*Model Answer\*\***: A circular diagram shows female frogs laying jelly eggs in water, which hatch into swimming tadpoles with gills and tails. Arrows show the tadpoles developing legs and lungs to become froglets, finally growing into tailless adult frogs on land. |

| **\*\*CH11-PIC-022\*\*** | **\*\*Model Answer\*\***:

1. Jelly-like egg clump floating in pond water.
2. Swimming tadpole with a long tail and gills.
3. Froglet with developed hind legs, lungs, and shrinking tail on a lily pad.
4. Tailless adult frog hopping on green land grass. |

| **\*\*CH11-PIC-023\*\*** | **\*\*Model Answer\*\***: It clearly contrasts the aquatic stage (tail, gills, water dweller) with the terrestrial stage (four legs, lungs, land dweller), visually proving how metamorphosis completely transforms a frog's body and habitat. |

| **\*\*CH11-PIC-024\*\*** | **\*\*Model Answer\*\***:

**\*\*Panel 1\*\***: **\*\* "I swim in the pond with my long tail and breathe with gills!" \*\***

**\*\*Panel 2\*\***: **\*\* "Look at my new legs and lungs! Soon my tail will be gone and I'll hop on land!" \*\*** |

| **\*\*CH11-PIC-025\*\*** | **\*\*Model Answer\*\***: The cover should feature a bright 4-stage circular lifecycle diagram, a close-up of a tadpole swimming in pond water, a froglet on a lily pad, a green adult frog catching an insect, and bold scientific ICSE typography. |